

Human CD1c-autoreactive T-cells recognise *Mycobacterium tuberculosis*-infected antigen-presenting cells and display cytotoxic effector programmes

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eLife Assessment

The study investigates how CD1c-restricted T cells respond to Mtb-infected APCs, leading to increased cytokine production and cytotoxic activity that may help control Mtb infection. The work is **important** and will interest researchers in the field. The supporting evidence is **solid**.

Abstract Tuberculosis (TB), caused by *Mycobacterium tuberculosis* (Mtb), remains the leading cause of death from infection globally yet the contribution of non-classical T-cell pathways to human immunity remains poorly defined. CD1c-autoreactive T-cells, which recognise self-lipids presented by the antigen-presenting molecule CD1c, are frequent in human blood but their role during infection remains unclear. Here, we investigate how CD1c-expressing antigen-presenting cells (APCs) and Mtb infection shape CD1c-autoreactive T-cell responses using engineered human APC systems, complemented by single-cell transcriptomic profiling to define the ex vivo phenotypic landscape of these T-cells. CD1c is present within human TB granulomas, whereas Mtb down-modulates CD1c expression on infected APCs, consistent with an immune evasion strategy. CD1c-autoreactive T-cells respond more strongly to Mtb-infected CD1c⁺ APCs than to uninfected cells, exhibiting enhanced activation, cytotoxicity, and diverse cytokine secretion via CD1c-dependent recognition. Under in vitro conditions, these T-cells reduce relative Mtb burden in infected phagocytes. Single-cell RNA sequencing reveals cytotoxic effector-memory programmes and expression of antimicrobial molecules, providing a mechanistic basis for these responses. Together, these findings define a human CD1c-restricted T-cell response to Mtb-infected APCs and identify autoreactive CD1c-restricted T-cells as a candidate cellular axis for lipid-directed immunity in TB.

Introduction

Tuberculosis (TB), caused by *Mycobacterium tuberculosis* (Mtb), is a major global epidemic (Wallis et al., 2016; World Health Organization, 2021; Hui et al., 2026). The only licensed vaccine, Bacille Calmette-Guérin (BCG), has variable efficacy against adult pulmonary disease, which drives transmission, and therefore new vaccines are urgently needed (Nguipdop-Djomo et al., 2016). Typically, novel TB vaccines have focussed on eliciting conventional T-cell responses to peptide antigens presented by highly polymorphic major histocompatibility complex (MHC) molecules (O'Garra et al., 2013; Jasenosky et al., 2015). Although progress has been made in developing new immunisation tools, one of the most promising candidates, M72, based on immunogenic peptides, achieved only 50% efficacy against TB (Ndiaye et al., 2015; Tameris et al., 2013; Schrager et al., 2020; Tait et al., 2019). This partial efficacy highlights the need to explore alternative approaches, including non-peptide antigens, to achieve the level of protection required for global TB control. Consequently, there is growing interest in harnessing unconventional T-cell responses as a complementary or alternative strategy (Joosten et al., 2019; Nunes-Alves et al., 2014). Generally, unconventional T-cells recognise antigens bound to non-polymorphic antigen-presenting molecules and therefore are not restricted by genetic variability. This group includes MR1-restricted mucosal-associated invariant T-cells, CD1d-restricted invariant natural killer T-cells (iNKT), HLA-E-, and CD1 group 1-restricted T-cells (Joosten et al., 2019; Ruibal et al., 2021).

The CD1 family (CD1a, CD1b, CD1c, CD1d) are non-polymorphic MHC class I-like proteins that bind and present self or foreign lipid antigens to $\alpha\beta$ and $\gamma\delta$ T-cells (Chancellor et al., 2018). CD1c is the most ubiquitously expressed group 1 CD1 molecule, present on antigen-presenting cells (APCs) such as dendritic cells (DCs), foamy macrophages, and B-cells (Mori et al., 2016; Melián et al., 1999). Several studies suggest a role for CD1c in host immunity to Mtb. CD1c presents Mtb-derived lipids to T-cells, including mannosyl- β 1-phosphomycoketide and phosphomycoketide (PM) (de Jong et al., 2007; Moody et al., 2000; Matsunaga et al., 2004; Scharf et al., 2010; Roy et al., 2014). CD1c-restricted and Mtb lipid-specific T-cells expand in the circulation of TB patients, and CD1c-PM-loaded tetramers detect PM-specific T-cells in individuals with latent TB (Moody et al., 2000; Ly et al., 2013).

However, it is notable that the majority of CD1c-restricted T-cells exhibit autoreactivity against CD1c expressing APCs in the absence of exogenous microbial antigens, suggesting recognition of CD1c bound to endogenous self-lipids (de Lalla et al., 2011; de Jong et al., 2010). These CD1c-autoreactive T-cells recognise diverse self-lipids, including phospholipids, cholesteryl-esters, methyllysophosphatidic acids (mLPAs), and monoacylglycerols (Wun et al., 2018; Mansour et al., 2016; Lepore et al., 2014). CD1c-autoreactive T-cells are relatively abundant in the circulation of healthy adults and are proposed to become activated in response to host lipids in autoimmune disease and cancer (de Lalla et al., 2011). CD1c-autoreactive T-cells isolated from patients with systemic lupus erythematosus provide help to CD1c⁺ B-cells, enhancing the production of IgG and contributing to immunopathology (Sieling et al., 2000). CD1c-autoreactive T-cells infiltrate lesions in Hashimoto's thyroiditis and Graves' disease, contributing to tissue destruction (Roura-Mir et al., 2005). CD1c-autoreactive T-cells recognise tumour-associated self-lipid antigens (mLPA) derived from leukaemic cells (Lepore et al., 2014). mLPA-specific T-cells efficiently kill CD1c⁺ acute leukaemia cells and protect immunodeficient mice against CD1c⁺ human leukaemia cells (Lepore et al., 2014). Therefore, CD1c-autoreactive T-cells contribute to the immune response in human autoimmune disease and tumour immunosurveillance. However, historically these diseases will not have exerted the dominant evolutionary pressure relative to mortality from infection.

Humans have co-evolved with bacteria, including gut commensals (Suzuki et al., 2022) and environmental and pathogenic mycobacteria (Brites and Gagneux, 2015). This long-standing relationship with microorganisms may have exerted evolutionary pressure on CD1c-reactive T-cells to contribute to immune defence against pathogenic organisms (Brites and Gagneux, 2015). Nevertheless, although CD1c-autoreactive T-cells are well described in human blood and have been implicated in autoimmunity and cancer, their role during infection remains poorly defined. This is an important knowledge gap because CD1c is positioned to present infection-altered lipid antigens, and autoreactive CD1-restricted T-cells may respond not only to steady-state self-lipids but also to infection-induced changes in lipid antigen presentation. Indeed, CD1c-autoreactive T-cells expand after Mtb infection of CD1 transgenic mice (hCD1Tg) in vivo (Felio et al., 2009). Furthermore, autoreactive T-cells restricted by other CD1 family members also exhibit enhanced responses to TLR agonists, mycobacteria, and

mycobacterial lipids, thereby indicating that these T-cells may also be activated during infection (*Li et al., 2011; Van Rhijn et al., 2016*). However, whether human CD1c-autoreactive T-cells directly respond to Mtb-infected APCs, whether this response is CD1c-TCR-dependent, and whether these cells mediate cytotoxic or antimicrobial effector functions has not been established.

Here, we define the response of human CD1c-autoreactive T-cells to Mtb-infected APCs using complementary human in vitro systems. We first examine CD1c expression in human TB lung tissue and determine how Mtb infection affects CD1c expression on primary DCs and engineered THP1-CD1c APCs. We then use CD1c-expressing APCs to expand and validate CD1c-autoreactive T-cell lines, and test their activation, cytokine secretion, cytotoxicity, and capacity to reduce relative Mtb burden in infected phagocytes. To establish whether enhanced recognition of infected APCs can be mediated through the TCR, we clone dominant CD1c-reactive TCRs and test their function in a Jurkat expression system. Finally, we use single-cell transcriptomic profiling of ex vivo CD1c-autoreactive T-cells to define their phenotypic landscape and effector programmes. Together, these data identify a previously unrecognised human CD1c-autoreactive T-cell response to Mtb-infected APCs and support these cells as candidate contributors to lipid-directed immunity during TB infection.

Results

Circulating CD1c-autoreactive T-cells in healthy donors

CD1c is an MHC class I-like molecule consisting of a heavy chain and $\beta 2m$ light chain (*Mansour et al., 2016*). CD1c-autoreactive T-cells are defined by their recognition of CD1c in the absence of exogenous lipid antigen (*de Jong et al., 2010; Wun et al., 2018; Lepore et al., 2014; Cotton et al., 2018*). To investigate the presence of CD1c-autoreactive T-cells in the blood of healthy donors, we developed novel in vitro assays to allow their specific expansion and subsequent detection by flow cytometry. To provide a controlled CD1c-expressing APC system while minimising conventional MHC-restricted T-cell activation, we generated THP1 knockout cells lacking both $\beta 2m$ and the Class II transactivator (CIITA). Loss of $\beta 2m$ removes surface expression of $\beta 2m$ -dependent molecules, including classical MHC class I and endogenous CD1 proteins, while CIITA knockout prevents MHC class II expression. Flow cytometric analysis confirmed that WT THP1 cells expressed $\beta 2m$ and classical MHC class I, with low detectable MHC class II and HLA-E (**Figure 1A**). In contrast, THP1-KO cells lacked detectable $\beta 2m$, MHC class I, MHC class II, HLA-E, CD1b, and CD1c. To provide specific CD1c stimulation, THP1-KO cells were transduced with a CD1c- $\beta 2m$ fusion construct to generate THP1-CD1c APCs. These cells expressed robust surface CD1c without detectable classical and non-classical MHC molecules (**Figure 1A**). Thus, the engineered THP1-CD1c system provides CD1c expression in the absence of detectable surface MHC-I or MHC-II, supporting its use to assess CD1c-mediated T-cell responses.

Purified CD3⁺ T-cells isolated from 14 healthy donors were labelled with the proliferation marker CellTrace Violet (CTV) and co-cultured with irradiated THP1-CD1c APCs without exogenously added lipid antigens. After 12 days in vitro culture, a fraction of cells had undergone cell division, as indicated by dilution of CTV, consistent with the presence of autoreactive T-cells (**Figure 1—figure supplement 1A**). To confirm this, we performed an activation-induced marker (AIM) assay and observed that a significant fraction of proliferating cells expressed the AIMs CD69, CD25, and CD137 upon re-challenge with THP1-CD1c cells (**Figure 1B and C**), with donor-to-donor variation in expansion. Importantly, CD1c-autoreactive T-cells expanded in most donors, with increased expression of CD69, CD25, and CD137 after overnight stimulation with THP1-CD1c cells (**Figure 1C**). The absence of activation in response to THP1-KO cells, together with the lack of detectable MHC-I and MHC-II on THP1-CD1c APCs, supports CD1c-dependent rather than conventional MHC-restricted activation. We performed a Luminex assay on short-term cultures expanded by THP1-CD1c APCs to measure cytokine secretion, studying proinflammatory and anti-inflammatory cytokines known to be critical in anti-mycobacterial immunity (*O'Garra et al., 2013*). CD1c-mediated stimulation, without exogenously added antigen, induced production of several cytokines, including IL-1 β , IL-2, IL-10, IL-13, and GM-CSF, suggesting a polyfunctional phenotype (**Figure 1—figure supplement 1B**). Thus, CD1c-autoreactive T-cells can be detected in healthy donors and expanded in short-term in vitro culture.

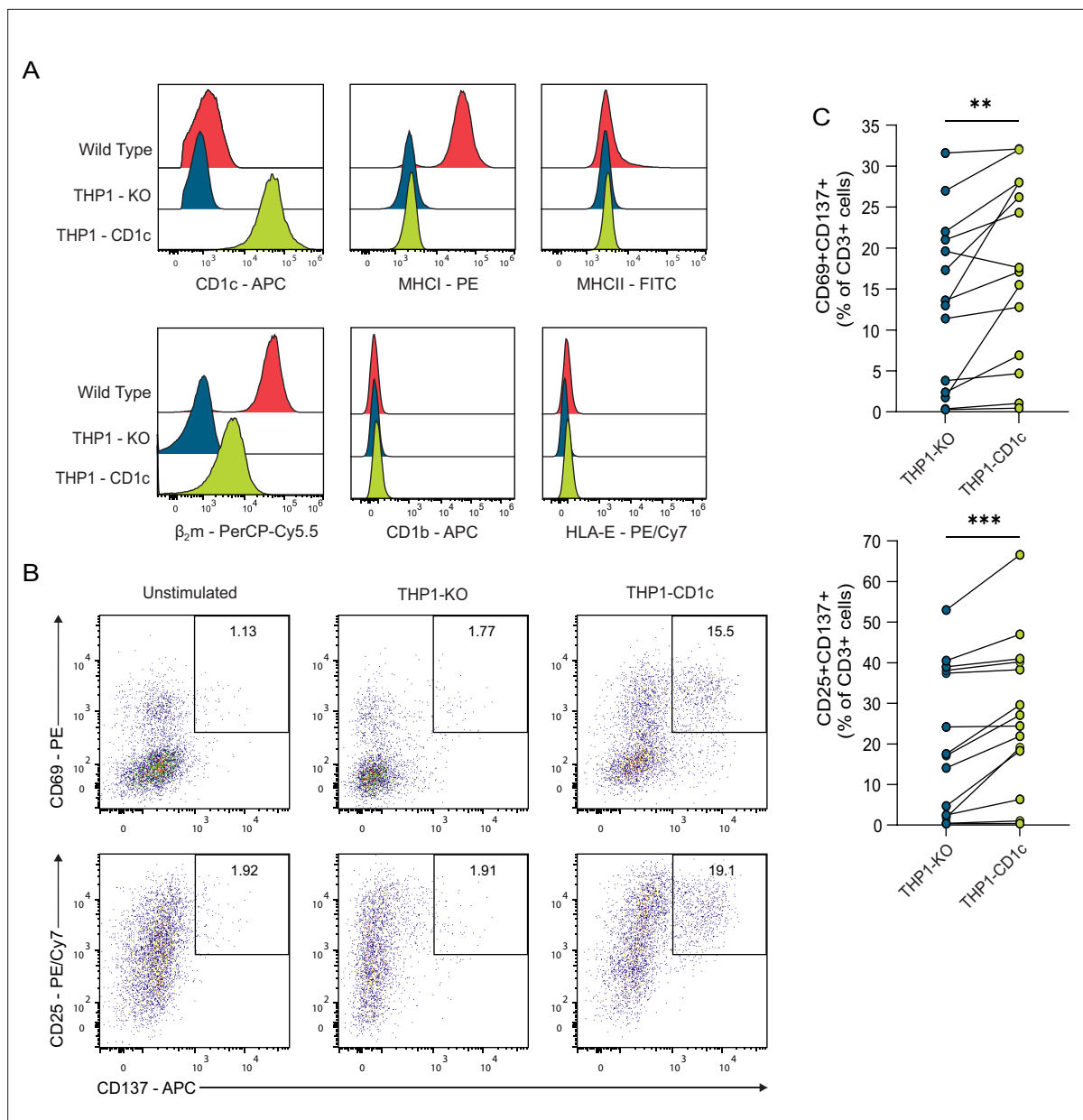


Figure 1. Autoreactive T-cells expand upon CD1c stimulation. **(A)** Representative histogram overlays showing CD1c, major histocompatibility complex (MHC) -I, MHC-II, β_2 -microglobulin (β_2m), CD1b, and HLA-E expression on wild-type THP1 cells, and engineered THP1-KO and THP1-CD1c antigen-presenting cells (APCs). **(B)** Representative flow cytometry analysis of T-cells from a healthy donor expanded with THP1-CD1c cells in the absence of exogenous lipid antigen and then cultured overnight with THP1 cells. Cells were gated on live CD3⁺CTV⁺ T-cells, and T-cell activation was measured with anti-CD69, anti-CD25, and anti-CD137. Plots show expression of CD69 and CD137 (top), CD25 and CD137 (bottom). Significant numbers of CD1c-autoreactive T-cells were present in expanded cultures. **(C)** Cumulative data from 14 healthy donors showing frequency of CD3⁺CTV⁺CD69⁺CD137⁺ T-cells (top), and CD3⁺CTV⁺CD25⁺CD137⁺ T-cells (bottom) from lines first expanded with THP1-CD1c and then overnight culture with THP1-KO or THP1-CD1c APCs. CD1c-autoreactive T-cells were present in the majority of donors. ** $p < 0.01$; *** $p < 0.001$ (C, Wilcoxon matched pairs signed-rank test).

The online version of this article includes the following figure supplement(s) for figure 1:

Figure supplement 1. CD1c-dependent activation and cytokine production by CD1c-autoreactive T-cells.

CD1c expression in human TB granulomas

We hypothesised that CD1c-autoreactive T-cells may have a role in mycobacterial infection, and so first investigated CD1c expression at the site of TB infection, which has not been previously characterised. We performed immunohistochemical staining of lung biopsies taken from patients with active pulmonary TB with a recombinant anti-CD1c antibody. CD1c expression was observed across

lung biopsies from all five TB patients, although staining was spatially heterogeneous rather than uniformly distributed throughout granulomatous lesions (**Figure 2A**). Panels 2Ai–iii show granulomatous regions, including central granuloma areas where CD1c staining was infrequent. In contrast, stronger CD1c staining was observed in distal inflammatory tissue remote from the granuloma centre and in lymphoid/B-cell follicle-rich regions (**Figure 2Av–vi**). Therefore, these data confirm the presence of CD1c-expressing cells in human TB lung tissue, while indicating that CD1c expression is spatially restricted and most prominent away from the granuloma centre.

Given the relatively low frequency of CD1c staining in central granuloma regions, we next investigated the dynamics of CD1c expression during Mtb infection. Previous reports suggested that Mtb regulates the antigen-presenting functions of DCs by downregulating CD1c expression, thereby repressing their response to infection (**Stenger et al., 1998b; Wen et al., 2016**). To investigate the impact of Mtb infection on CD1c expression, we first interrogated an RNA sequencing dataset derived from five healthy donors where monocyte-derived DC (MoDC) were infected with live Mtb (**Pacis et al., 2019**). At 48 hr after Mtb infection, CD1 group 1 expression was significantly reduced in infected MoDCs derived from all five donors relative to uninfected cells (**Figure 2B**). Next, we validated these results at the protein level on MoDCs derived from healthy donors by flow cytometry. Our results corroborate the transcriptomic dataset and reveal significantly reduced expression of CD1c protein on primary MoDCs after 48 hr of live Mtb infection measured by flow cytometry (**Figure 2C** and **Figure 2—figure supplement 1A**). To investigate whether Mtb regulates CD1c expression on our engineered THP1-CD1c APCs, we measured CD1c expression after Mtb infection. Contrary to the reduced expression of CD1c on infected primary MoDCs, CD1c expression on THP1-CD1c APCs did not change following Mtb infection (**Figure 2D** and **Figure 2—figure supplement 1B**). This likely reflects the engineered nature of the THP1-CD1c system, in which CD1c is expressed from a CD1c- β 2m fusion construct rather than the endogenous CD1C locus. Thus, this model allows CD1c-dependent T-cell responses to Mtb-infected APCs to be assessed without the confounding effect of infection-induced CD1c loss. This was important for subsequent functional experiments, because primary DCs infected with Mtb would be expected to lose endogenous CD1c expression, making it difficult to distinguish reduced T-cell activation caused by CD1c downregulation from changes in CD1c-presented antigens. Overall, our results reveal spatially heterogeneous CD1c expression in human TB lung tissue, with staining most apparent away from the granuloma centre, and show that Mtb downregulates CD1c on primary DCs but not in the engineered THP1-CD1c APC model used for functional assays.

CD1c-autoreactive T-cells are cytotoxic

To investigate CD1c-autoreactive T-cell function in more detail, we derived highly enriched CD1c-autoreactive T-cell lines from the peripheral blood of healthy donors using two independent approaches. First, we expanded T-cells from a donor that exhibited robust proliferation in response to THP1-CD1c APCs (**Figure 1B**), followed by CD1c-endo tetramer-guided flow cytometric sorting and in vitro expansion. In parallel, we enriched blood-derived T-cells from an independent donor using CD1c-endo streptamers, followed by CD1c-endo dextramer sorting and expansion (**Figure 3—figure supplement 1**). The gating strategy, sorting approach, and post-expansion validation are shown in **Figure 3—figure supplement 1**. After expansion, sorted cells showed strong staining with CD1c-endo tetramers, whereas unstained cells and an irrelevant tetramer showed no detectable staining, confirming specific enrichment of CD1c-endo-binding T-cells. The resulting donor-derived T-cell lines showed near-uniform CD1c-endo tetramer staining after expansion, consistent with highly efficient enrichment of CD1c-endo-binding T-cells (**Figure 3A and C**). Both lines expressed $\alpha\beta$ TCR and CD4, with no detectable $\gamma\delta$ TCR or CD8 expression (**Figure 3B and D**). Functionally, both lines were activated in response to THP1-CD1c APCs but not parental THP1-KO APCs, as shown by upregulation of CD69 and CD25 (**Figure 3E**). Together, these data demonstrate that these T-cell lines are highly enriched, CD1c-endo tetramer-positive, $\alpha\beta$ TCR+CD4+ T-cell populations with CD1c-dependent functional reactivity.

CD1c-autoreactive T-cells have been reported as having cytotoxic activity (**Lepore et al., 2014; Spada et al., 2000**), and so we next investigated their cytotoxicity against target cells stimulated with UV-killed Mtb. We first cultured CD1c-autoreactive T-cells with THP1-CD1c APCs treated with increasing doses of UV-killed Mtb and measured cell death. CD1c-autoreactive T-cells mediated

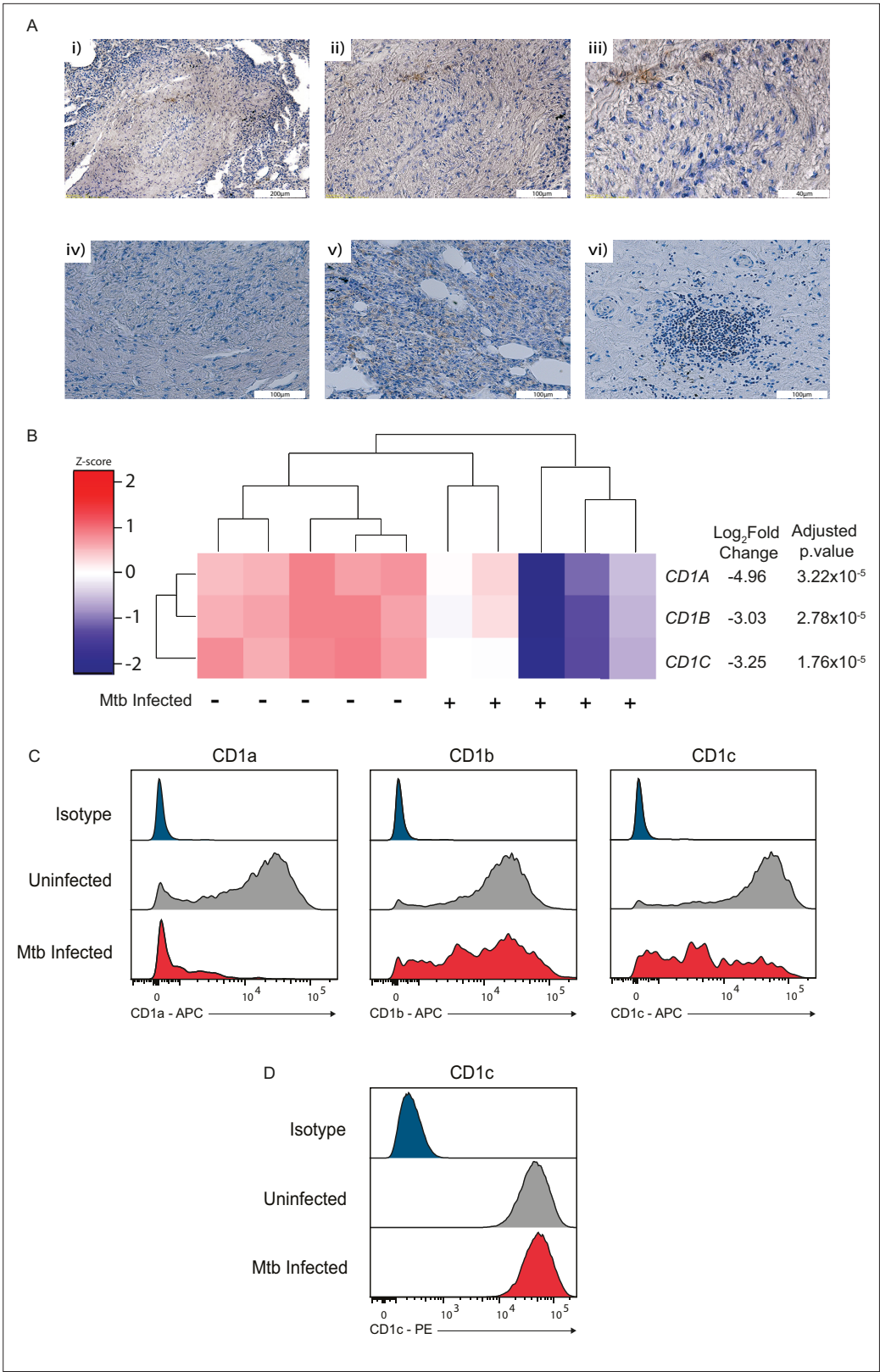


Figure 2. CD1c is expressed in human tuberculosis (TB) and downregulated in antigen-presenting cells (APCs) by *M. tuberculosis* (Mtb) infection. **(A)** Lung biopsies from patients with active TB were stained with anti-CD1c antibodies. (i–iii) Representative granulomatous regions, including central granuloma areas where CD1c immunoreactivity is infrequent. (iv) Control staining of the same granuloma with secondary antibody only and

Figure 2 continued on next page

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avidin biotin-peroxidase complex (ABC) detection shows no immunoreactivity. (v) CD1c staining is more apparent in inflammatory tissue remote from the TB granuloma centre. (vi) CD1c staining in B-cell follicles adjacent to the granuloma. (B) RNA-seq heatmap showing significant reduction in CD1A, CD1B, and CD1C expression in monocyte-derived dendritic cells (MoDCs) from five donors at 48 hr after Mtb infection (MOI = 1). Differential gene expression was performed on filtered normalised counts using the voom-limma pipeline in RStudio, with p-value adjustment performed using the Benjamini-Hochberg method. Differentially expressed genes were identified as having $\log_2FC > +1$ (upregulated) or < -1 (downregulated) with adjusted p-values < 0.05 . (C) MoDC histograms of normalised counts demonstrating a reduction in CD1a, CD1b, and CD1c expression on differentiated MoDCs following live Mtb infection (MOI = 1). Data representative of an experiment conducted in two donors performed in triplicate. (D) Flow cytometry histograms showing CD1c expression on THP1-CD1c cells at 72 hr following infection with live Mtb (MOI = 1). Mtb infection does not change CD1c expression on THP1-CD1c cells. Data representative of two experiments performed in triplicate.

The online version of this article includes the following figure supplement(s) for figure 2:

Figure supplement 1. Mtb infection alters CD1 molecule expression on antigen-presenting cells.

cytotoxicity towards THP1-CD1c APCs treated with UV-killed Mtb, in a Mtb dose-dependent manner, while no cytotoxicity occurred against untreated THP1-CD1c APCs (**Figure 3F**). Irrelevant control T-cells did not induce notable cytotoxicity against THP1-CD1c APCs (**Figure 3F**). In addition, CD1c-autoreactive T-cells induced significant cytotoxicity, in a T-cell dose-dependent manner, against THP1-CD1c APCs treated with UV-killed Mtb (MOI = 10), but not to untreated controls (**Figure 3G**). Increasing the dose of the control T-cells did not induce cytotoxicity (**Figure 3G**). These results demonstrate that CD1c-autoreactive T-cells are cytotoxic towards CD1c⁺ APCs treated with UV-killed Mtb.

CD1c-autoreactive T-cells exhibit enhanced lysis of Mtb-infected APCs beyond baseline autoreactivity

To study T-cell cytotoxicity in more detail, we performed complementary assays studying live Mtb infection. We cultured T-cells with THP1-KO and THP1-CD1c that were either uninfected or infected with live Mtb. First, we investigated T-cell activity by measuring the activation markers CD69 and CD25 by flow cytometry. CD1c-autoreactive T-cells were activated when cultured with uninfected THP1-CD1c APCs but not with control THP1-KO APCs, consistent with CD1c-dependent autoreactivity (**Figure 4A and B**). Notably, increased activation of CD1c-autoreactive T-cells occurred when they were cultured with Mtb-infected THP1-CD1c APCs (**Figure 4A and B**). To investigate T-cell cytotoxicity more directly, we cultured the T-cells with APCs before measuring target-cell viability by flow cytometry (**Figure 4C**). Importantly, we observed lysis of Mtb-infected THP1-CD1c APCs by CD1c-autoreactive T-cells, greater than the lysis of uninfected THP1-CD1c APCs (**Figure 4D**).

Recent studies suggest that activation of APCs with TLR agonists can enhance the activation of CD1-autoreactive T-cells (Li et al., 2011; Brennan et al., 2011). Therefore, to investigate whether the enhanced cytotoxicity of CD1c-autoreactive T-cells was facilitated via TLR-mediated mechanisms, we cultured T-cells with THP1-CD1c APCs that were infected with Mtb or stimulated with the TLR agonists Pam3Cys (TLR2) or LPS (TLR4). TLR agonists did not enhance T-cell cytotoxicity above the autoreactive response, whereas Mtb infection of APCs induced significant cytotoxicity of THP1-CD1c APCs (**Figure 4E**). Overall, our results confirm that CD1c-autoreactive T-cells kill Mtb-infected CD1c⁺ APCs, which is not replicated by TLR activation alone.

The enhanced Mtb reactivity is mediated via CD1c recognition by the TCR

To provide further mechanistic insight, we performed single-cell TCR sequencing of CD1c-autoreactive T-cells (**Figure 3A**). Single CD1c-endo tetramer-positive T-cells were sorted into individual wells for targeted TCR sequencing. After filtering and manual curation, 11 single cells yielded productive paired $\alpha\beta$ TCR sequences. The repertoire was oligoclonal, with two dominant productive clonotypes accounting for 10 of 11 paired TCRs. The dominant clonotype, designated EM1, was detected in 6 of 11 cells and consisted of an alpha chain containing TRAV4 and TRAJ5 paired with a beta chain containing TRBV7-8, TRBD1, and TRBJ2-7. The second clonotype, designated EM2, was detected in 4 of 11 cells and consisted of an alpha chain containing TRAV1-2 and TRAJ26 paired with a beta chain

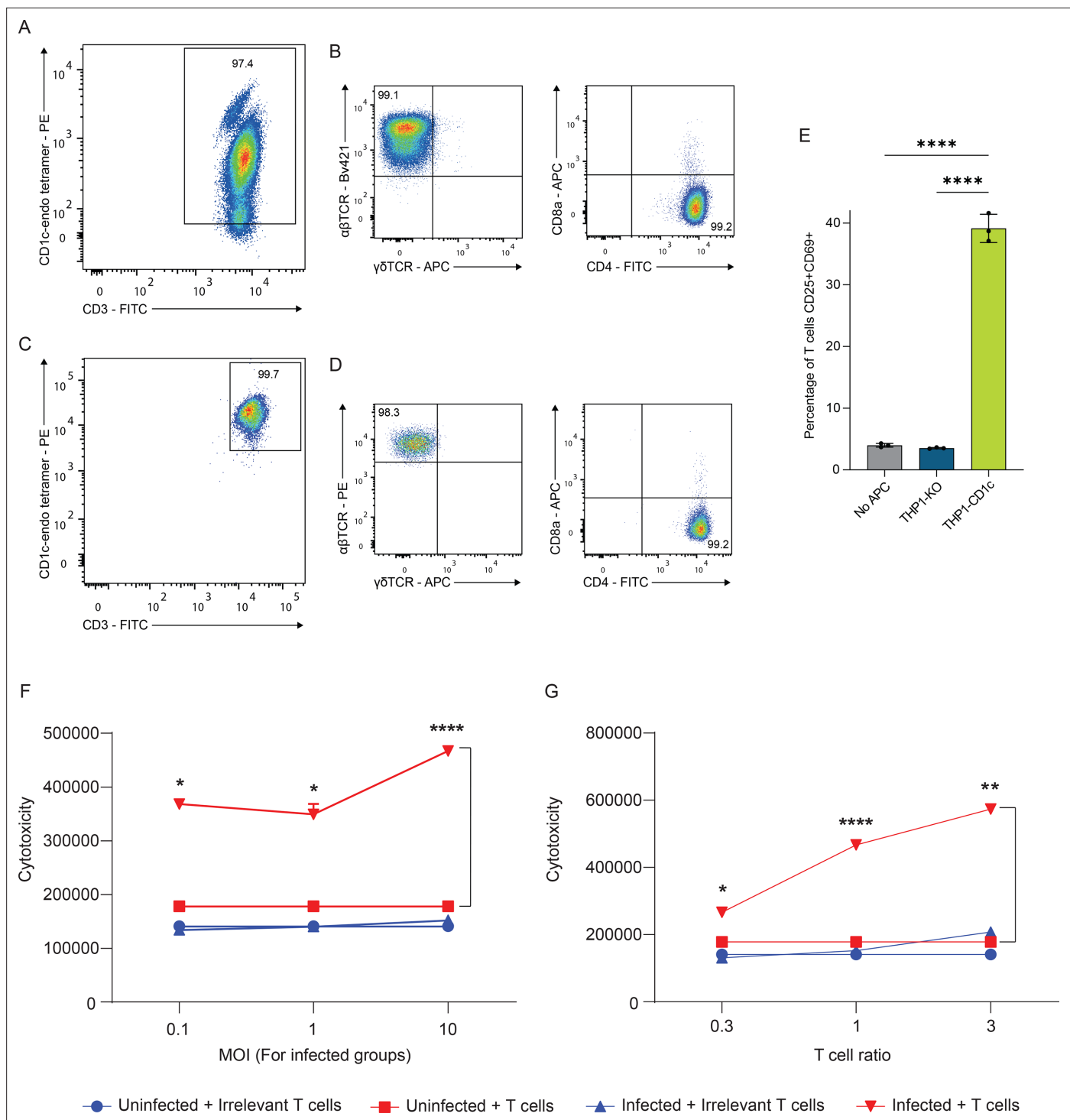


Figure 3. CD1c-autoreactive T-cells are cytotoxic to *M. tuberculosis* (Mtb)-stimulated cells. (A–D) CD1c-endo tetramer staining and phenotypic characterisation of T-cell lines generated from two healthy donors. (A) T-cells generated from peripheral blood mononuclear cells (PBMCs) after CD1c-endo streptamer enrichment, CD1c-endo dextramer flow cytometric sorting, and subsequent in vitro expansion (Figure 3—figure supplement 1). (B) Phenotypic analysis demonstrating that the cells in (A) are $\alpha\beta$ TCR⁺ and CD4⁺. (C) T-cells generated following expansion with THP1-CD1c cells, CD1c-endo tetramer-guided sorting, and subsequent in vitro expansion. (D) Phenotypic analysis demonstrating that the cells in (C) are $\alpha\beta$ TCR⁺ and CD4⁺. (E) T-cell lines are activated in response to THP1-CD1c antigen-presenting cells (APCs). Rested T-cells and T-cells cultured with THP1-KO APCs served as control. T-cell activity was determined by measuring the upregulation of CD69 and CD25 by flow cytometry. Data is representative of three experiments from the two donor-derived lines performed in triplicate. (F) CD1c-autoreactive T-cells display significant cytotoxicity in a dose-dependent manner against THP1-CD1c APCs treated with increasing doses of UV-killed Mtb. The untreated/no-infection condition represents MOI = 0. The indicated MOI

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values refer only to THP1-CD1c APCs treated with UV-killed Mtb. No cytotoxicity was observed by T-cells cultured with untreated THP1-CD1c APCs or irrelevant control T-cells, lacking CD1c restriction. (G) CD1c-autoreactive T-cells displayed significant cytotoxicity in a T-cell dose-dependent manner against UV-killed Mtb-treated THP1-CD1c APCs. No cytotoxicity was observed by T-cells cultured with untreated THP1-CD1c APCs or with irrelevant control T-cells. Cytotoxicity was measured using a ToxiLight assay. Data is representative of two independent experiments, each performed in triplicate. * $p < 0.05$; ** $p < 0.01$; **** $p < 0.0001$ (E, one-way ANOVA with Tukey's multiple comparison test; F–G, two-way ANOVA).

The online version of this article includes the following figure supplement(s) for figure 3:

Figure supplement 1. Flow cytometry gating strategy depicting live CD3⁺ T-cells comprising the negative (cells that did not bind the streptamers) or the CD1c-endo streptamer positive T-cell fraction (containing 1.14% CD1c-endo-positive T-cells) stained with CD1c-endo dextramers.

containing TRBV2, TRBD1, and TRBJ1-1 (**Figure 5A**). We utilised the $\beta 2m$ knockout JRT3.5 Jurkat T-cell line, which lacks both TCR and CD1c expression, to avoid self-reactivity by Jurkat T-cells in functional studies (**Figure 5—figure supplement 1**). This reporter system was used to test whether EM1 and EM2 confer CD1c-dependent recognition and activation, rather than cytotoxic effector function. We transduced the $\beta 2m$ knockout JRT3.5 Jurkat T-cell line with the EM1 and EM2 TCRs. Staining of TCR-transduced Jurkats with CD1c-endo tetramers showed clear binding of EM1 and EM2 Jurkat T-cells, consistent with direct engagement of CD1c by the TCR (**Figure 5B**). CD1c-endo tetramers did not stain the control parental Jurkat T-cells (**Figure 5B**). In addition, culturing the Jurkat T-cells on plates coated with CD1c-endo protein resulted in the activation of both EM1 and EM2 Jurkat T-cell lines demonstrated by upregulation of CD69, suggesting TCR-dependent signalling via CD1c engagement (**Figure 5C**).

Mtb infection downregulates endogenous CD1c on primary MoDCs (**Figure 2B and C**), and therefore we tested EM1 and EM2 TCR reactivity using the engineered THP1-CD1c system, in which CD1c expression is preserved during infection (**Figure 2D**). We observed Jurkat T-cell line activation through upregulation of CD69 when cultured with uninfected THP1-CD1c APCs but not with control THP1-KO APCs, confirming the autoreactivity of the TCRs (**Figure 5D**). Importantly, we observed significantly more Jurkat T-cell activation when cells were cultured with Mtb-infected THP1-CD1c APCs (**Figure 5D**). These data suggest that the enhanced activation of CD1c-autoreactive T-cells against Mtb-infected target cells is mediated by TCR recognition of CD1c.

CD1c-autoreactive T-cells release cytokines associated with TB immunity and suppress Mtb luminescence

To further investigate T-cell function in the context of Mtb infection, we cultured CD1c-autoreactive T-cells with APCs that were treated with UV-killed Mtb and analysed cytokine secretion. CD1c-autoreactive T-cell cytokine secretion was increased somewhat in response to untreated THP1-CD1c APCs, further corroborating their CD1c-autoreactivity (**Figure 6—figure supplement 1A**). Critically though, CD1c-autoreactive T-cells secreted higher levels of cytokines in response to Mtb-treated THP1-CD1c APCs (**Figure 6A and B**). No cytokines were released by an irrelevant control T-cell line (**Figure 6A**), and the response was CD1c-mediated, with no upregulation in response to THP1-KO cells (**Figure 6—figure supplement 1B**). To determine cytokines released by the APCs alone, we treated THP1-CD1c cells with Mtb and measured their cytokine release. Mtb-treated THP1-CD1c APCs released significant amounts of IL-8 and RANTES, but not any other cytokine (**Figure 6—figure supplement 2**).

CD1c-autoreactive T-cells secreted the proinflammatory cytokines IFN- γ , TNF- α , IL-1 α , IL-2, and GM-CSF in response to Mtb-treated APCs, in a dose-responsive effect (**Figure 6A and B**). Additionally, CD1c-autoreactive T-cells produced IL-4, IL-5, IL-10, and IL-13, demonstrating polyfunctionality. Finally, we assessed whether CD1c-autoreactive T-cells could restrict Mtb growth in phagocytes using luminescent live Mtb. Following infection, extracellular bacteria were removed by washing before T-cell co-culture, and luminescence was then monitored as a measure of viable lux-expressing Mtb. These T-cells marginally reduced Mtb growth in THP1-KO APCs, likely due to basal cytokine secretion (Reljic et al., 2010). Importantly, luminescence was significantly further reduced in infected THP1-CD1c APCs, consistent with a CD1c-dependent reduction of relative Mtb burden, as luminescence closely correlates with colony-forming units (Bielecka et al., 2017; **Figure 6C**). Because this assay measures luminescence of lux-expressing Mtb and does not distinguish intracellular from extracellular bacteria, we interpret these data as evidence of reduced net viable bacterial burden under in vitro

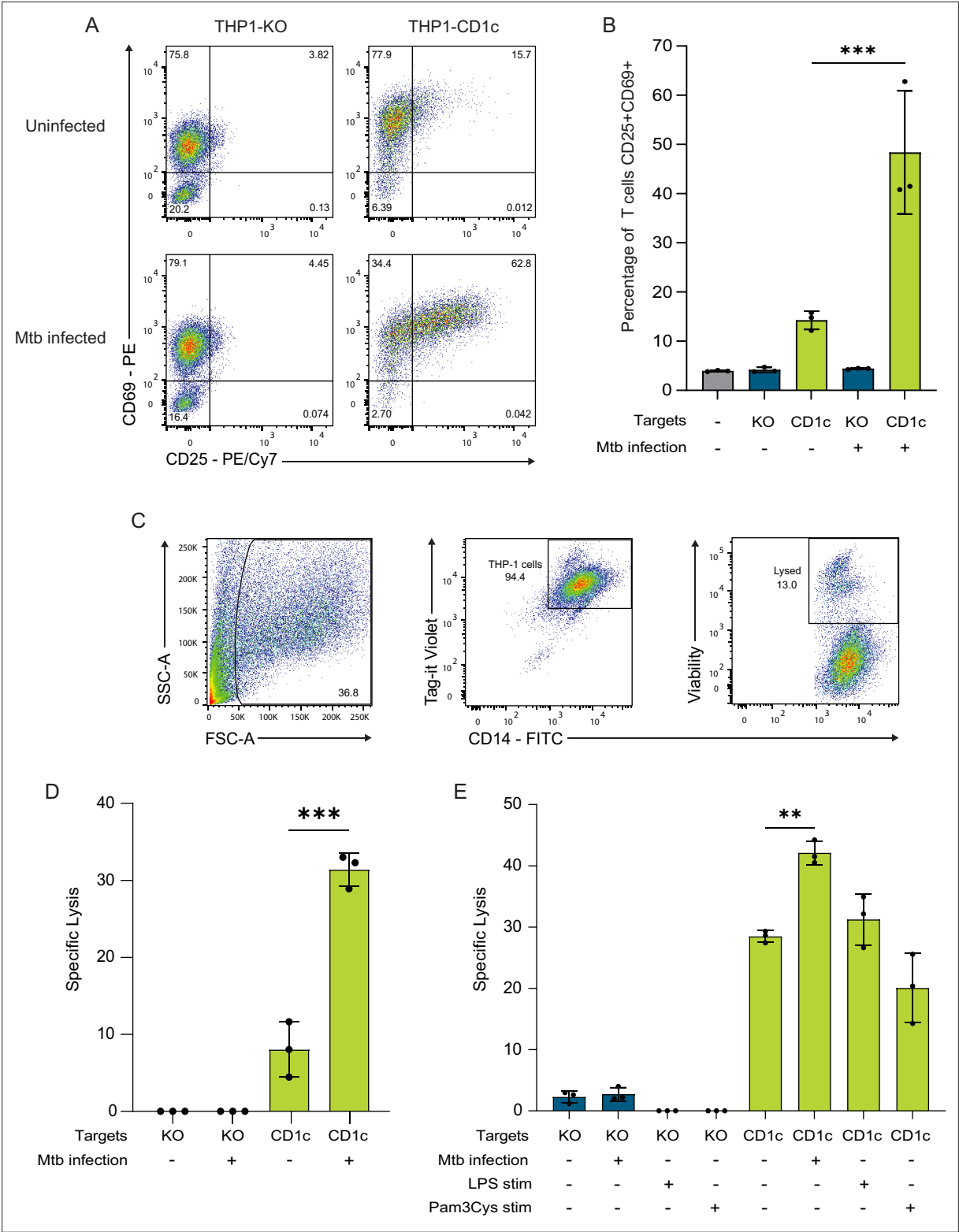


Figure 4. CD1c-autoreactive T-cells lyse target cells infected with live *M. tuberculosis* (Mtb). **(A)** Flow cytometry dot plots and **(B)** bar graphs showing CD1c-autoreactive T-cells are activated by THP1-CD1c antigen-presenting cells (APCs) but not THP1-KO APCs, with significantly greater activation when THP1-CD1c APCs are infected with live Mtb (MOI = 1). **(C)** Gating strategy for measuring the T-cell-mediated lysis of THP1 cells. Prior to T-cell culture, THP1 cells were stained with Tag-it Violet to allow identification of THP1 target cells in co-culture. LIVE/DEAD Fixable Near-IR Dead Cell Marker was used to measure the proportion of dead THP1 cells. Matched APC-only controls, including uninfected and Mtb-infected THP1 cells cultured without

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T-cells, were included in parallel to define baseline target-cell death for each condition. (D) CD1c-autoreactive T-cells lyse THP1-CD1c APCs but not THP1-KO APCs. Killing is significantly enhanced when THP1-CD1c cells are infected with live Mtb (MOI = 1). Specific lysis was calculated by subtracting baseline target-cell death observed in matched APC-only controls from target-cell death observed in the corresponding T-cell co-culture condition. (E) Lysis of THP1-CD1c is increased with Mtb infection, but not enhanced with LPS or Pam3CSK4 stimulation. Specific lysis was calculated relative to matched APC-only controls cultured in parallel. Data are representative of three independent experiments, each performed in triplicate. ** $p < 0.01$; *** $p < 0.001$ (B, D, and E, one-way ANOVA with Tukey's multiple comparison test).

co-culture conditions, rather than selective intracellular bacterial killing. Overall, our findings demonstrate that CD1c-autoreactive T-cell responses are significantly greater towards Mtb-infected APCs than the basal autoreactive response and reduce relative Mtb burden under these in vitro conditions.

CD1c-autoreactive T-cells exhibit cytotoxic effector-memory phenotype

Given their ability to restrict Mtb growth and secrete proinflammatory cytokines in response to infected APCs, we performed unbiased transcriptomic profiling to gain deeper insight into the functional properties of CD1c-autoreactive T-cells. To this end, we isolated T-cells ex vivo from two donors using barcoded 'dCODE' CD1c-endo dextramers and performed single-cell RNA sequencing (scRNA-seq). CD3⁺ T-cells were enriched from peripheral blood mononuclear cells (PBMCs), and dextramer-positive and -negative T-cells were sorted for comparative analysis (Figure 7A, Figure 7—figure supplement 1). UMAP visualisation of the combined dataset revealed diverse T-cell populations, including naïve and effector-memory (TEM) CD4⁺ and CD8⁺ T-cells (Figure 7B, Figure 7—figure supplement 2).

We focused on high-confidence CD1c-autoreactive T-cells, defined by co-positivity for CD1c-endo dextramer and dCODE barcode, and identified 2117 such cells across both donors (Figure 7—figure supplement 3). These cells were predominantly enriched in cytotoxic CD4⁺ and CD8⁺ effector-memory clusters (Figure 7C and D), although they also displayed transcriptional diversity spanning multiple T-cell states present in the broader dataset. Phenotypic analysis showed high expression of cytotoxic mediators, including *GZMA*, *GZMK*, and *GNLY*, as well as effector molecules such as *CCL5* and *NKG7* (Figure 7E). Pathway enrichment analysis revealed significant upregulation of gene sets related to response to other organisms, T-cell-mediated cytotoxicity, and cell killing (Figure 7F). Together, these findings indicate that CD1c-autoreactive T-cells exhibit cytotoxic effector-memory phenotypes and express key antimicrobial effector molecules, providing mechanistic insight into their potential effector function against Mtb-infected APCs.

Discussion

The majority of human CD1c-restricted T-cells are activated by exposure to CD1-expressing APCs in the absence of foreign lipid antigens (de Lalla et al., 2011; de Jong et al., 2010). Indeed, such CD1c-autoreactive T-cells comprise 2% of all circulating human $\alpha\beta$ T-cells (de Lalla et al., 2011), and studies have suggested a functional role for these T-cells in human autoimmune disease (Roura-Mir et al., 2005) and cancer (Lepore et al., 2014). However, for almost the entirety of human evolution, infection is likely to have exerted a greater selective pressure affecting survival and reproduction than autoimmune disease or cancer (Karlsson et al., 2014; Fumagalli et al., 2011). Here, we found that polyclonal CD1c-autoreactive T-cells produced higher levels of Th1 and Th2 cytokines when stimulated with Mtb-infected APCs relative to uninfected cells, mediated increased cytotoxicity against Mtb-infected APCs in a CD1c-mediated manner, and reduced the relative viable Mtb burden under in vitro co-culture conditions (Bielecka et al., 2017). To gain insight into the cellular programmes associated with this antimicrobial effector potential, we performed scRNA-seq of ex vivo isolated CD1c-autoreactive T-cells from healthy donors. These cells were transcriptionally diverse but predominantly exhibited cytotoxic effector-memory phenotypes. Notably, they expressed high levels of cytolytic molecules, including *GZMA*, *GZMK*, and *GNLY*, as well as effector mediators such as *CCL5* and *NKG7*, consistent with a role in direct target-cell killing (Stenger et al., 1998a). Granulysin is one of the few T-cell-derived molecules with demonstrated antimicrobial activity against Mtb (Stenger et al., 1998a), yet few human T-cell subsets express it in this context. Combined with our finding that CD1c-autoreactive T-cells can reduce Mtb luminescence in vitro, this places them among a rare population with antimicrobial activity in this experimental setting. These findings provide a

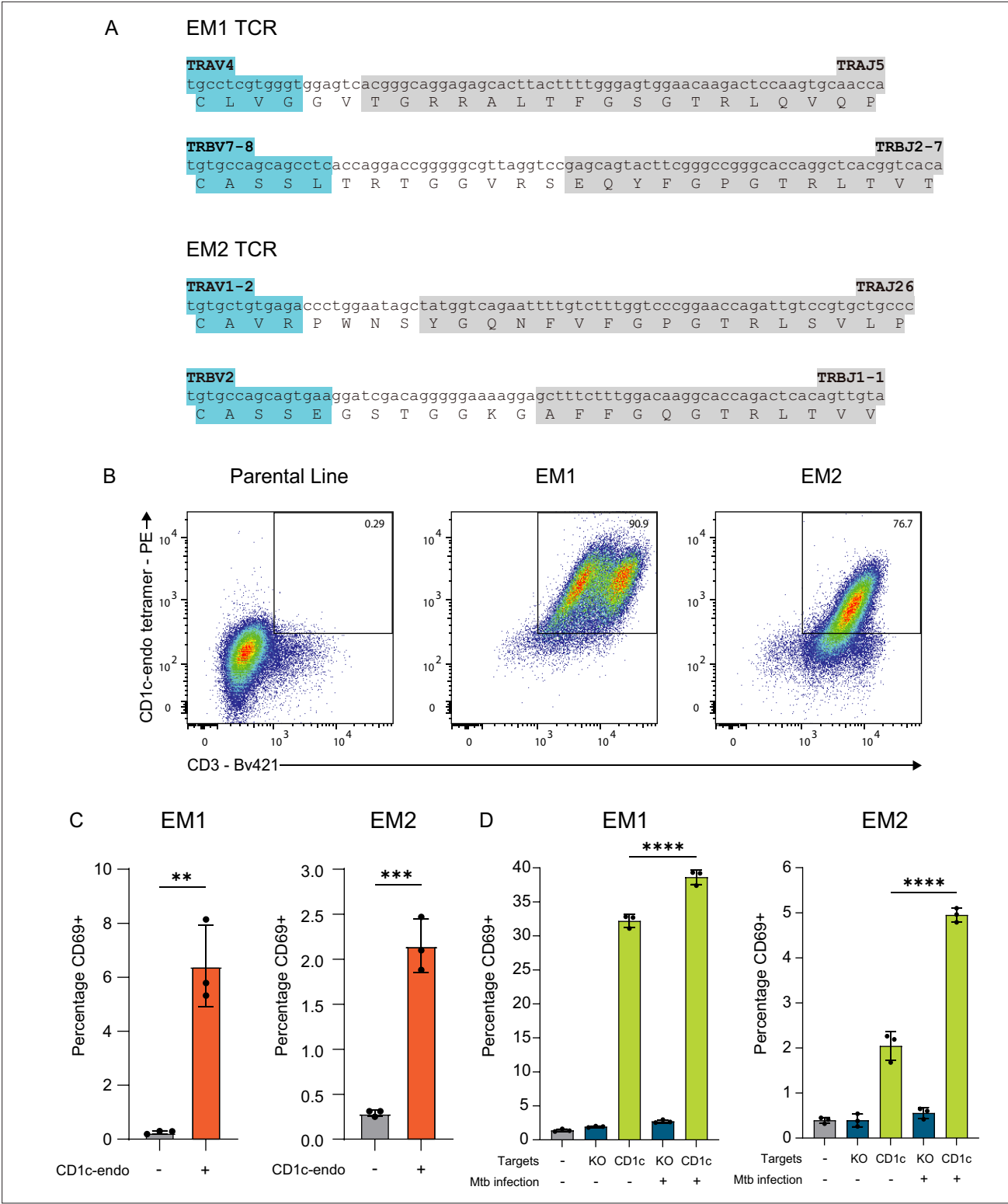


Figure 5. TCR-CD1c interactions mediate the response to *M. tuberculosis* (Mtb) infection. **(A)** The alpha (top) and the beta (bottom) chain sequences of EM1 and EM2 TCRs. Variable region (blue), N additions (white), and joining segment (grey) are shown. After filtering and manual curation, 11 single cells yielded productive paired $\alpha\beta$ TCR sequences. EM1 and EM2 represented 6/11 cells and 4/11 cells, respectively. **(B)** CD1c-endo tetramer staining of parental Jurkats and Jurkat T-cells transduced to express EM1 and EM2 TCRs. **(C)** Percentage activation of Jurkat T-cells transduced with EM1 and EM2 TCRs in response to uncoated or CD1c-endo coated wells. T-cell activity was measured by flow cytometry staining with anti-CD69. **(D)** Jurkat T-cells transduced with EM1 and EM2 TCRs are activated by THP1-CD1c antigen-presenting cells (APCs) but not THP1-KO APCs, with significantly greater

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activation when THP1-CD1c APCs are infected with live Mtb (MOI = 1). Data are representative of two independent experiments, each performed in triplicate. ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$ (C, unpaired t-test; D, one-way ANOVA).

The online version of this article includes the following figure supplement(s) for figure 5:

Figure supplement 1. Validation of CD1c and TCR expression in JRT3.5 Jurkat T-cell lines.

mechanistic explanation for how CD1c-autoreactive T-cells could mediate cytotoxicity and contribute to control of intracellular Mtb, supporting their classification as functionally distinct effector T-cells rather than merely autoreactive bystanders. To our knowledge, this is the first single-cell transcriptional analysis of human CD1c-autoreactive T-cells, offering preliminary insight into their phenotype and potential functional diversity. Our data suggest that CD1c-responsive T-cells, previously described as 'autoreactive', respond more robustly to Mtb-infected APCs, raising the possibility that, in addition to roles in autoimmunity and cancer, these cells may also contribute to antimicrobial responses. Together, these findings address a previously unresolved question: whether human CD1c-autoreactive T-cells, known mainly in the context of autoimmunity and cancer, can functionally respond to Mtb-infected APCs. By integrating engineered APC systems, TCR transfer, cytotoxicity assays, and ex vivo single-cell transcriptomics, we show that these cells exhibit CD1c-dependent activation, cytotoxicity, cytokine production, and antimicrobial activity in response to Mtb-infected APCs. This links CD1c-autoreactivity to infection-associated effector function and supports these cells as candidate contributors to lipid-directed immunity in TB.

Two previous studies utilising either in vitro activation-based assays or CD1c-endo tetramers have confirmed the presence of CD1c-autoreactive T-cells in human peripheral blood (de Lalla et al., 2011; Gherardin et al., 2021). Although there was a discrepancy in their estimation of ex vivo frequencies for CD1c-autoreactive T-cells, both studies showed that these T-cells are a natural component of the human T-cell pool (de Lalla et al., 2011; Gherardin et al., 2021). Moreover, both showed that CD1c-autoreactive T-cells were enriched for CD4⁺ or DN T-cells, and exhibited memory and adaptive T-cell features closely aligned with conventional MHC-restricted T-cells (de Lalla et al., 2011; Gherardin et al., 2021). CD1c-autoreactive T-cells expressed diverse TCRs, including $\alpha\beta$, $\gamma\delta$, and $\delta/\alpha\beta$ (Gherardin et al., 2021). Although the TCR repertoire was generally diverse, they were enriched for the TRAV17, TRAV38-1, and TRBV4-1 gene segments (Gherardin et al., 2021; Guo et al., 2018). Functionally, CD1c-autoreactive T-cell clones exhibit a Th1- or a Th0-like phenotype, releasing IFN- γ , TNF- α , GM-CSF, IL-4, and IL-5 (de Lalla et al., 2011). Corroborating these earlier studies, we detected CD1c-autoreactive T-cells in the blood of healthy donors, and demonstrated that CD1c-autoreactive T-cells were polyfunctional. The variable expansion observed across donors likely reflects heterogeneity in CD1c-autoreactive precursor frequency, TCR repertoire composition, and activation state, as expected for primary human autoreactive T-cell populations (de Lalla et al., 2011). The functional and single-cell datasets presented capture different levels of this repertoire. The functional assays used two donor-derived CD1c-autoreactive T-cell lines that were CD4⁺ $\alpha\beta$ TCR⁺, whereas ex vivo scRNA-seq revealed broader CD1c-endo-binding populations, including both CD4⁺ and CD8⁺ cytotoxic effector-memory T-cells. These approaches are therefore complementary: the T-cell lines provide mechanistic evidence of CD1c-dependent function, while single-cell profiling captures the wider phenotypic diversity of the CD1c-autoreactive compartment.

We generated CD1c-restricted $\alpha\beta$ T-cell lines with a baseline autoreactive response that was significantly greater when responding to Mtb-infected cells, with increased activation markers, cytotoxicity, and cytokine secretion, and an associated reduction in Mtb luminescence in infected APCs. The Jurkat T-cell experiments with EM1 and EM2 TCRs support a key role for the TCR in mediating response through direct recognition of CD1c. Because Jurkat cells are not cytotoxic effector cells (Grailer et al., 2023), these experiments were designed to test TCR-dependent recognition and activation rather than target-cell killing. These findings suggest that changes in CD1c expression or cytokine secretion alone are unlikely to explain the enhanced T-cell activity observed towards Mtb-infected cells and instead point to a role for lipid antigen presentation on CD1c. Our findings align with a study in 2005 where autoreactive CD1 group 1 restricted T-cell clones were generated by limited dilution, after stimulation of CD4-depleted T-cells with MoDCs pulsed with hydrophobic microbial extracts derived from Mtb, *Escherichia coli*, and *Yersinia enterocolitica* (Vincent et al., 2005). A panel of 15 autoreactive clones were generated with the majority being CD1a- and CD1b-restricted and only

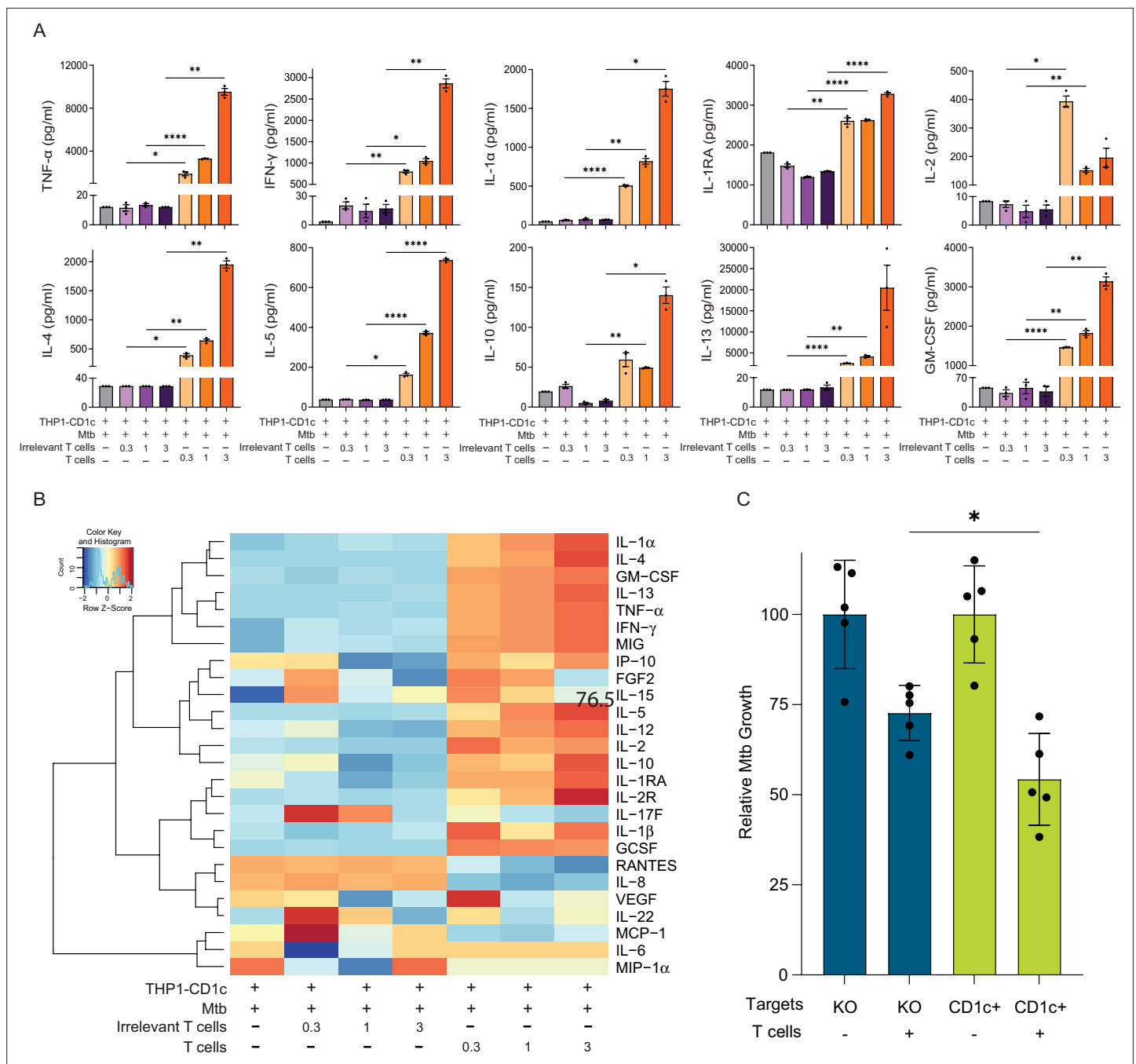


Figure 6. CD1c-autoreactive T-cells secrete diverse cytokines and reduce *M. tuberculosis* (Mtb) luminescence. **(A)** Cytokines secreted by CD1c-autoreactive T-cells cultured with UV-killed Mtb-treated THP1-CD1c antigen-presenting cells (APCs). CD1c-autoreactive T-cells produced significant amounts of the Th1 cytokines TNF- α , IFN- γ , IL-1 α , and GM-CSF and the Th2 cytokines IL-4, IL-5, IL-10, and IL-13 in a T-cell dose-dependent manner. Irrelevant control T-cells did not release cytokines. **(B)** Heatmap summarising cytokines released by CD1c-autoreactive T-cells or irrelevant control T-cells (CD1c unrestricted), in response to UV-killed Mtb-treated THP1-CD1c APCs. Red indicates high concentrations, and blue indicates low concentrations. Cytokine secretion was measured using a Luminex assay. Data are representative of two independent experiments, each performed in triplicate. **(C)** CD1c-autoreactive T-cells reduce Mtb luminescence significantly when cultured with THP1-CD1c APCs relative to when they are cultured with THP1-KO APCs. T-cells cause some reduction in THP1-KO cells, and this is greater in THP1-CD1c cells. * $p < 0.05$; ** $p < 0.01$; **** $p < 0.0001$ (A, two-way ANOVA; C, unpaired t-test).

The online version of this article includes the following figure supplement(s) for figure 6:

Figure supplement 1. Mtb enhances CD1c-dependent cytokine responses by CD1c-autoreactive T-cells.

Figure supplement 2. UV *M. tuberculosis* (Mtb)-treated THP1-CD1c antigen-presenting cells (APCs) secrete chemokines IL-8 and RANTES (pink bars), in comparison to media-only (black bars) and untreated THP1-CD1c APCs (blue bars).

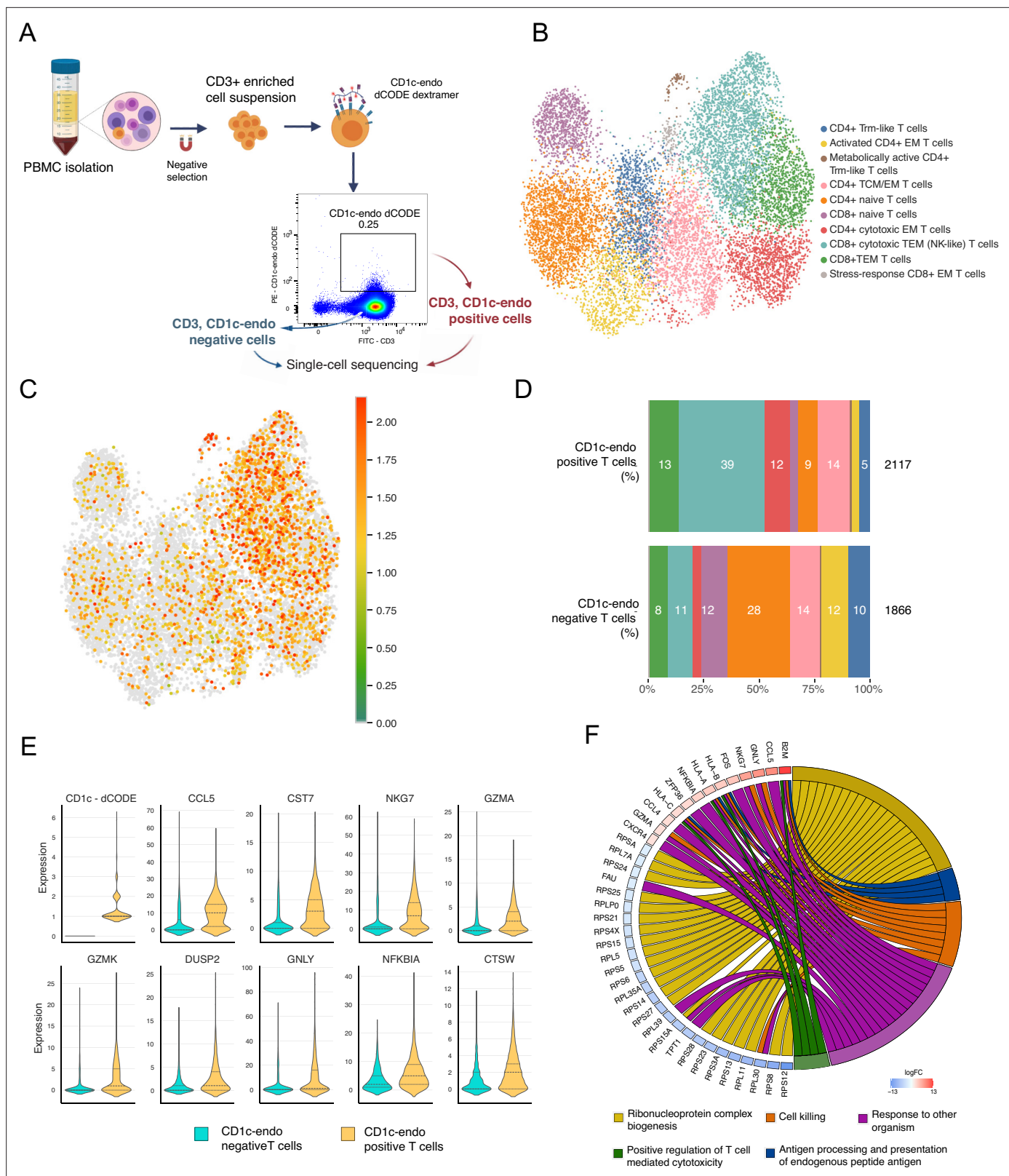


Figure 7. Single-cell profiling reveals that CD1c-endo dextramer positive T-cells are enriched for cytotoxic and effector phenotypes. **(A)** Schematic overview of the sample processing workflow. Peripheral blood mononuclear cells (PBMCs) were isolated from two donors and CD3⁺ T-cells were subsequently enriched by negative selection. Cells were stained with CD1c-endo dCODE dextramer, and CD3⁺CD1c-endo⁺ (positive) and CD3⁺CD1c-endo⁻ (negative) T-cells were sorted for single-cell RNA sequencing. **(B)** UMAP visualisation of 11,804 single T-cells clustered by transcriptional profile.

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Figure 7 continued

Clusters were annotated as functional T-cell subsets, including CD4⁺ and CD8⁺ naïve, central memory (TCM), effector memory (EM and TEM), cytotoxic, and stress-response populations. Additional subsets included metabolically active T-cells and tissue-resident memory-like (Trm-like) CD4⁺ cells. TRM: tissue-resident memory; EM: effector memory; TEM: T effector memory; TCM: T central memory. (C) UMAP projection showing CD1c-endo dextramer binding intensity, reflecting the number of bound dextramer molecules per cell. Cells with higher binding intensities are enriched within cytotoxic and effector CD4⁺ and CD8⁺ subsets. (D) Bar plots showing the proportional distribution of T-cell subtypes among CD1c-endo-positive (top) and -negative (bottom) populations. CD1c-endo-positive T-cells were enriched for cytotoxic and effector subsets. (E) Violin plots showing expression levels of the top 10 differentially expressed genes by Wilcoxon rank-sum test. CD1c-endo-positive T-cells upregulate genes associated with cytotoxicity (e.g. NKG7, GZMA, GZMK, GNLY, CTSW) and inflammation (e.g. CCL5, NFKBIA, DUSP2), consistent with a distinct effector phenotype. (F) Chord plot illustrating functional enrichment of differentially expressed genes in CD1c-endo-positive T-cells. Upregulated genes are associated with biological processes such as cell killing, antigen processing, and presentation and response to other organisms. In contrast, downregulated genes predominantly map to ribonucleoprotein complex biogenesis. The colour gradient indicates the log fold-change (logFC) in gene expression.

The online version of this article includes the following figure supplement(s) for figure 7:

Figure supplement 1. Identification and isolation of CD1c-endo-reactive T-cell populations.

Figure supplement 2. Dot plot showing the top 5 ranked marker genes for each annotated T-cell population, derived from all high-quality T-cells (n=11,804).

Figure supplement 3. Filtering strategy for identifying CD1c-endo dextramer-bound T-cells in single-cell RNA sequencing (scRNA-seq) data from sorted populations.

two clones being CD1c-autoreactive (Vincent et al., 2005). While the clones exhibited weak auto-reactivity, the response to microbial antigens far exceeded the self-reactive response seen in assays of T-cell proliferation (Vincent et al., 2005), similar to our findings and suggesting an antimicrobial role. TCR transfer experiments supported a role for the TCRs in promoting this dual reactivity to self and foreign antigen. However, further data from clone Mt2.33, which was a CD1c-autoreactive clone generated through Mtb antigen stimulation, was not provided. Further evidence for CD1c-restricted dual reactivity comes from a study identifying Vδ1⁺ γδ T-cells using CD1c-PM tetramers, where Jurkat cells transduced with these γδ TCRs exhibited spontaneous activation and stronger responses to microbial lipid antigens, again implicating TCR-driven recognition of CD1c-presented ligands (Roy et al., 2016). Another study investigated the CD1b-autoreactive clone HJ1, which was isolated from hCD1Tg mice, in *Listeria monocytogenes* infection (Li et al., 2011). While HJ1 T-cells exhibited auto-reactivity towards CD1b, responses were enhanced by treatment of APCs with *Listeria* or the TLR agonists Pam3Cys (TLR2) and LPS (TLR4) (Li et al., 2011). Together, these observations suggest that some autoreactive CD1-responsive T-cells may have evolved as anti-microbial effectors, but when studied experimentally without such stimuli, a residual autoreactive phenotype is observed.

TLR activation during infection of APCs promotes the accumulation of stimulatory self-lipid antigens, contributing to the enhanced activation of iNKT cells (Brennan et al., 2011). Indeed, treatment of APCs with LPS or bacteria induced the accumulation of stimulatory iNKT self-lipid agonists b-d-glucopyranosylceramides (b-GlcCer) (Brennan et al., 2011). In our system, Mtb infection of THP1-CD1c APCs enhanced cytotoxicity by CD1c-autoreactive T-cells, but this effect was not reproduced by TLR2 or TLR4 agonists alone. This suggests that generic TLR2/4-mediated APC activation is insufficient to explain the enhanced response to Mtb-infected cells. Together with the TCR transfer experiments, these data point towards a CD1c-dependent signal that is increased or altered during Mtb infection. The molecular identity of this signal remains unresolved. One possibility is that Mtb infection alters the CD1c-presented lipid repertoire, either through presentation of Mtb-derived lipids (Vincent et al., 2005; Roy et al., 2016), infection-induced host 'stress lipids' (Mansour et al., 2016), or bacterial and mammalian shared lipids similar to those described for CD1b-autoreactive T-cells (Van Rhijn et al., 2016; Reinink et al., 2019). Alternatively, infection may alter lipid processing, trafficking or loading onto CD1c, increasing presentation of permissive self-lipids that can sit deeply within the CD1c groove and expose the CD1c surface for TCR recognition (Wun et al., 2018). Stimulatory lipids could include cholesteryl esters (Mansour et al., 2016) or other infection-induced lipid species. Thus, the additional signal provided by Mtb-infected THP1-CD1c cells is likely to be a qualitative or quantitative change in CD1c-associated lipid presentation, rather than CD1c expression alone or generic APC activation, a hypothesis supported by our TCR transfer experiments.

A limitation of this study is that we did not directly identify the lipids presented by CD1c on Mtb-infected APCs, making it unclear whether the relevant ligands are host-derived, bacterial, or shared lipid species. Future studies should isolate CD1c molecules from Mtb-infected APCs and use mass spectrometry-based lipidomics to define the lipid species associated with enhanced T-cell activation (*Huang et al., 2023*). Comparing these autoreactive populations with non-autoreactive CD1c-restricted T-cells that recognise defined microbial lipids will also be key, as the two may represent complementary modes of lipid immune surveillance. Although Mtb luminescence correlates closely with CFU counts in this system (*Bielecka et al., 2017*), it remains an indirect measure of viable bacterial burden, and future studies using CFU plating or complementary bacterial quantification methods will be important to confirm the extent of CD1c-dependent growth restriction. A further limitation is that the scRNA-seq analysis was performed on CD1c-endo-binding T-cells from two donors. Although subsampling analysis supported the robustness of the transcriptional patterns within this dataset, larger donor cohorts will be required to determine the consistency of cytotoxic effector-memory programmes across the broader human CD1c-autoreactive T-cell repertoire. In addition, our experiments rely on in vitro human cell systems, and in vivo validation of CD1c-restricted responses will benefit from specialised CD1-transgenic models that accommodate human CD1c biology, beyond what is possible in standard mouse systems (*Felio et al., 2009; Zhao et al., 2015*).

Our observation of CD1c expression within human lung TB granulomas provides further evidence for a potential role for lipid-specific T-cell immunity in the host immune response to Mtb. Our findings are consistent with reports suggesting that CD1c expression is targeted by Mtb and BCG, leading to reduced expression on APCs such as DCs (*Stenger et al., 1998b; Wen et al., 2016; Aquino et al., 2011; Gagliardi et al., 2004*). A recent study identified a potential mechanism, with Mtb driving expression of the microRNA miR-381-3p and thereby suppressing CD1c (*Wen et al., 2016*). This has important implications for interpreting functional assays using primary DCs. Although primary CD1c-expressing DCs are biologically relevant APCs, Mtb-induced CD1c downregulation would make it difficult to determine whether altered T-cell activation reflects changes in the CD1c-presented lipid repertoire or simply reduced CD1c at the cell surface. For this reason, the engineered THP1-CD1c system provides a controlled reductionist model in which CD1c expression is maintained during infection, allowing CD1c-TCR-dependent activation to be tested directly. This difference between primary MoDCs and THP1-CD1c APCs likely reflects the engineered nature of the THP1-CD1c system. Rather than regulation from the endogenous CD1C locus, CD1c is expressed from a heterologous CD1c- β 2m fusion construct lacking the 3'-untranslated region where miR-381-3p can bind to disrupt translation (*Wen et al., 2016*). Therefore, our data are consistent with the idea that CD1c-autoreactive responses may contribute to protective host immunity, and that Mtb-mediated suppression of CD1c expression on primary APCs may represent an immune evasion strategy (*Wen et al., 2016*). Future studies comparing TB lesions with healthy lung and non-TB inflammatory lung tissue will be important to determine whether the spatial pattern of CD1c expression observed here is specific to TB or reflects broader inflammatory remodelling of lung APC populations.

Taken together, our findings suggest that T-cells previously described as autoreactive may also play a role in anti-mycobacterial immunity, and that this phenotype could have been shaped by evolutionary adaptation to mycobacterial exposure. Mtb has killed many more humans than any other single infection, and humans and Mtb may have co-evolved for 70,000 years (*Comas et al., 2013; Paulson, 2013*), demonstrating the significant impact mycobacteria will have had on human immune development (*Reichmann et al., 2026*). We identify functional responses to Mtb-infected cells by CD1c-autoreactive T-cells, including activation, cytotoxicity, cytokine secretion, and reduction in Mtb luminescence in vitro, that clearly exceed the autoreactive phenotype. Together, our findings suggest that CD1c-autoreactive T-cells may contribute to immune responses against infection and warrant further investigation in the context of vaccine development.

Methods

Key resources table

Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Strain, strain background (<i>Mycobacterium tuberculosis</i>)	H37Rv	DOI: https://doi.org/10.7554/eLife.52668		Live Mtb used for APC infection (MOI = 1).
Strain, strain background (<i>Mycobacterium tuberculosis</i>)	H37Rv lux	DOI: https://doi.org/10.7554/eLife.52668		Live Mtb used for APC infection (MOI = 1).
Cell line (<i>Homo sapiens</i>)	THP1 WT	ATCC	RRID:CVCL_0006	Myelomonocytic leukaemia line. Identity/phenotype verified in the study by flow cytometry and STR profiling.
Cell line (<i>Homo sapiens</i>)	THP1-KO	This paper		THP1 derivative generated by CRISPR-Cas9 knockout of B2M and CIITA.
Cell line (<i>Homo sapiens</i>)	THP1-CD1c	This paper		THP1-KO derivative transduced with a beta2m-CD1c single-chain construct and sorted for CD1c expression.
Cell line (<i>Homo sapiens</i>)	J.RT3-T3.5 Jurkat; beta2m-knockout J.RT3-T3.5 Jurkat	Pierre Vantourout, King's College London		TCR-deficient Jurkat system; beta2m-knockout cells were a kind gift from Pierre Vantourout (King's College London). Expected phenotype verified by flow cytometry and STR profiling.
Cell line (<i>Homo sapiens</i>)	HEK293TN	ATCC	RRID:CVCL_UL49	Used for lentiviral particle production.
Biological sample (<i>Homo sapiens</i>)	Peripheral blood mononuclear cells (PBMCs)/primary T-cells	Healthy blood-bank donors	13/SC/0043	PBMCs isolated by Ficoll-Paque; used for T-cell isolation, expansion, MoDC generation, and single-cell studies.
Biological sample (<i>Homo sapiens</i>)	Paraffin-embedded lung tissue from patients with tuberculosis	University Hospital Southampton histology archive	Ethics:12/NW/0794; SRB04_14	Surplus archived clinical tissue used for immunohistochemistry.
Antibody	T-cell phenotyping antibodies: anti-CD3-BV510/FITC (mouse monoclonal, UCHT1); anti-alpha/beta TCR-BV421 (mouse monoclonal, IP26); anti-gamma/delta TCR-APC (mouse monoclonal, B1); anti-CD4-FITC (mouse monoclonal, RPA-T4); anti-CD8-APC (mouse monoclonal, HIT8a)	BioLegend		5 µL per test.
Antibody	MHC/beta2m antibodies: anti-MHC-I-PE (mouse monoclonal, W6/32); anti-MHC-II-FITC (mouse monoclonal, Tu39); anti-beta2m-PerCP-Cy5.5 (mouse monoclonal, A17082A); anti-HLA-DR-PE/Cy7 (mouse monoclonal, L243)	BioLegend		5 µL per test.
Antibody	CD1 antibodies: anti-CD1a-APC (mouse monoclonal, HI149); anti-CD1b-APC (mouse monoclonal, SN13(K5-1B8)); anti-CD1c-APC/PE (mouse monoclonal, L161)	BioLegend		5 µL per test.
Antibody	Activation antibodies: anti-CD69-PE (mouse monoclonal, FN50); anti-CD25-PE/Cy7 (mouse monoclonal, BC96); anti-CD137-APC (mouse monoclonal, 4B4-1)	BioLegend		5 µL per test.

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Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Antibody	APC/myeloid markers: anti-CD14-FITC (mouse monoclonal, M5E2); anti-CD11c-BV421 (mouse monoclonal, Bu15); anti-CD36 (mouse monoclonal, 5–271)	BioLegend		5 µL per test.
Antibody	Anti-CD1c (rabbit recombinant monoclonal, EPR23189-196)	Abcam		Immunohistochemistry (1:500), overnight at 4°C.
Antibody	Goat anti-rabbit secondary antibody	2B Scientific		Immunohistochemistry (1:800).
Recombinant DNA reagent	beta2m-CD1c single-chain construct	This paper		Used to generate THP1-CD1c cells; beta2m fused to CD1c.
Recombinant DNA reagent	CD1c-beta2m single-chain expression cassette in pCDNA3.1 with Avi-tag/His6 and BirA	This paper		Used to express soluble human CD1c for tetramer/dextramer generation.
Recombinant DNA reagent	EM1 TCR in pELNS lentivector	GenScript		Full-length TCR generated from tetramer-guided TCR sequencing and used to transduce beta2m-knockout J.RT3-T3.5 cells.
Recombinant DNA reagent	EM2 TCR in pELNS lentivector	GenScript		Full-length TCR generated from tetramer-guided TCR sequencing and used to transduce beta2m-knockout J.RT3-T3.5 cells.
Recombinant DNA reagent	pCMV-VSV-G; pRSV.REV; pMDL.pg.RRE	Other		Lentiviral packaging plasmids used with pELNS. Source: University of Pennsylvania
Recombinant protein	CD1c-endo monomer	This paper		Soluble human CD1c-beta2m single-chain protein produced using the Expi293 expression system and purified by nickel affinity/size exclusion chromatography.
Recombinant protein	CD1c-endo tetramer	This paper		CD1c-endo monomer multimerised using PE-streptavidin (BioLegend).
Other	CD1c-endo dextramer/PE-dCODE dextramer	Immudex/this paper		CD1c-endo multimers generated using PE-labelled dextran backbones; dCODE dextrans used for single-cell sorting.
Other	CD1c-endo streptamer	This paper		Generated from CD1c-endo containing a Strep-Tag II sequence and Strep-Tactin magnetic microbeads (IBA Lifesciences).
Commercial assay or kit	EasySep Human T-cell Isolation Kit	STEMCELL Technologies		Used to purify pan T-cells from healthy donor blood.
Commercial assay or kit	CD14+ monocyte negative-selection magnetic beads	STEMCELL Technologies		Used to isolate CD14+ monocytes for MoDC generation; exact product identifier not stated in the manuscript.
Commercial assay or kit	Expi293 Expression System	Thermo Fisher Scientific		Used for recombinant soluble CD1c expression.
Commercial assay or kit	ToxiLight cytotoxicity assay	Lonza		Used to measure cytotoxicity against UV-killed Mtb-treated target cells.
Commercial assay or kit	Human cytokine 35-plex panel	Thermo Fisher Scientific		Multiplex cytokine analysis; exact catalogue identifier not stated in the manuscript.

Continued on next page

Continued

Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Commercial assay or kit	NGS library preparation kit	New England Biolabs		Used for targeted single-cell TCR sequencing; exact kit name/catalogue identifier not stated in the manuscript.
Commercial assay or kit	Chromium Single Cell V(D)J Reagent Kit v1.1; Chromium Chip G	10x Genomics		Used for single-cell library preparation.
Chemical compound, drug	Dasatinib	Axon		50 nM during tetramer/streptamer staining.
Chemical compound, drug	CellTrace Violet (CTV)	Thermo Fisher Scientific		Used to track T-cell proliferation.
Chemical compound, drug	LIVE/DEAD Fixable Near-IR Dead Cell Marker	Thermo Fisher Scientific		Used for viability discrimination in flow cytometry and single-cell sorting.
Chemical compound, drug	Tag-it Violet	BioLegend		Used to identify THP1 target cells in lysis assays.
Chemical compound, drug	Pam3CSK4	InvivoGen		TLR2 agonist used at 1 µg/mL in lysis assays.
Chemical compound, drug	LPS	Sigma		TLR4 agonist used at 1 µg/mL in lysis assays.
Peptide, recombinant protein	rhIL-2 (Proleukin)	Chiron		10 IU/mL during expansion cultures; 100 IU/mL during post-sort expansion.
Peptide, recombinant protein	rhGM-CSF; rhIL-4	Miltenyi Biotec		Used for MoDC differentiation (25 ng/mL rhGM-CSF; 20 ng/mL rhIL-4).
Software, algorithm	FlowJo	FlowJo LLC	v10.8.1 (RRID:SCR_008520)	Flow cytometry analysis.
Software, algorithm	GraphPad Prism	GraphPad Software, Inc	v10 (RRID:SCR_002798)	Statistical analysis.
Software, algorithm	Seven Bridges	Seven Bridges		Analysis of targeted TCR sequencing data; version not stated in manuscript.
Software, algorithm	Cell Ranger	10x Genomics	v7.2.0 (RRID:SCR_017344)	Demultiplexing/alignment of single-cell sequencing data to GRCh38 (2020A).
Software, algorithm	Scrublet; scDblFinder	Open-source software	RRID:SCR_018098	Independent doublet-detection methods used for scRNA-seq quality control; versions not stated.
Software, algorithm	enaBrowserTools; kallisto; tximport; edgeR; limma	Open-source software	enaBrowserTools v1.1.0; kallisto v0.46.1; tximport v1.28.0; edgeR v3.14.0; limma v3.56.2	RNA-seq retrieval, quantification, and differential expression analysis.

Flow cytometry

The following fluorescent reagents were used: All antibodies were from BioLegend unless stated; anti-CD3-BV510 (clone UCHT1); anti-CD3-FITC (clone UCHT1); anti-MHC-I-PE (clone W6/32); anti-MHC-II-FITC (clone Tu39); anti-β2m-PerCP-Cy5.5 (clone A17082A); anti-αβTCR-BV421 (clone IP26); anti-γδTCR-APC (clone B1); anti-CD4-FITC (clone RPA-T4); anti-CD8-APC (clone HIT8a); anti-CD14-FITC (clone M5E2); anti-CD11c-BV421 (clone Bu15); anti-HLA-DR-PE/Cy7 (clone L243); anti-CD1a-APC (clone HI149); anti-CD1b-APC (clone SN13(K5-1B8)); anti-CD1c-APC (L161); anti-CD1c-PE (clone L161); anti-CD69-PE (clone FN50); anti-CD25-PE/Cy7 (clone BC96); anti-CD137-APC (clone 4B4-1); LIVE/DEAD Fixable Near-IR Dead Cell Marker and CTV (both from Thermo Fisher Scientific); and propidium iodide (Sigma). T-cells were treated with 50 nM dasatinib (Axon) and blocked with Human TruStain FcX (BioLegend) for 30 min at 37°C, and anti-CD36 (clone 5-271, BioLegend) for 20 min before the addition of tetramers and staining reagents. Cells were stained for 45 min, extensively

washed and acquired on a FACS Aria Fusion, or a FACS Aria II (both from BD Biosciences). Data was analysed using FlowJo VX software (FlowJo LLC).

Cell lines

The following cell lines were utilised in this study: THP1 (myelomonocytic leukaemia), J.RT3-T3.5 (TCR β -deficient T-cell leukaemia), and HEK293TN (human embryonic kidney). THP1 cells were obtained from ATCC, HEK293TN cells were obtained from ATCC, and β 2m-knockout J.RT3-T3.5 Jurkat cells were kindly provided by Pierre Vantourout (King's College London). Cell-line identity and expected phenotype were verified by flow cytometry using appropriate cell-surface marker panels and by STR profiling. Cell cultures were routinely tested for mycoplasma contamination by polymerase chain reaction (PCR) and were negative. The engineered THP1-KO cells were generated using CRISPR-Cas9 technology to prevent the expression of β 2m and CIITA. Following knockout of β 2m and CIITA, the cells were transduced using a lentiviral system with a plasmid containing a β 2m-CD1c single-chain gene construct to generate the THP1-CD1c cell line. Three days post-transduction, CD1c⁺ cells were sorted by flow cytometry and maintained in culture. Jurkat, THP1, and HEK293TN cells were maintained in complete RPMI or DMEM (for HEK293TN) containing 10% foetal bovine serum (Sigma), 1% non-essential amino acids, 1% GlutaMAX, 1% sodium pyruvate, and 1% penicillin/streptomycin (all from Thermo Fisher Scientific). All primary cells used in this study were isolated from PBMCs obtained using Ficoll-Paque (Cytiva) from healthy blood-bank donors and maintained in T-cell medium comprising RPMI supplemented with 5% human AB serum (Merck), 1% penicillin/streptomycin, 1% GlutaMAX, 1% non-essential amino acids, and 1% sodium pyruvate (all from Thermo Fisher Scientific). Pan T-cells from healthy donors, purified using the EasySep Human T-cell Isolation Kit (STEMCELL Technologies), were labelled with CTV according to the manufacturer's instructions and stimulated with irradiated (80 Gy) THP1-KO or THP1-CD1c cells at a 6:1 ratio, or without THP1 cells as a control. THP1 cells were irradiated to prevent APC proliferation during the 12-day T-cell expansion culture while preserving their use as stimulatory cells. On days 4, 6, 8, 10, and 12 post-stimulation, 10 IU/mL rhIL-2 (Proleukin, Chiron) was added. At 12 days post-stimulation, cells were rechallenged with THP1-KO or THP1-CD1c APCs overnight before staining for activation-marker expression on live CD3⁺ CTV low cells by flow cytometry. Some cultures were stained with CD1c-endo tetramers directly on day 12, after which tetramer-positive T-cells were sorted by flow cytometry and expanded in the presence of phytohaemagglutinin (PHA; Thermo Fisher Scientific) (1 μ g/mL), rhIL-2 (100 U/mL), and irradiated PBMCs (5 $\times$ 10⁵ cells/mL).

Tetramers

Human CD1c was produced as a single chain construct with β 2m N-terminally fused to the heavy chain via a flexible glycine-serine linker. Expression cassettes were cloned in-frame with C-terminal Avi-tag and His6 tag in pCDNA3.1 encoding an additional BirA Ligase. Protein expression was performed using the Expi293 Expression System (Thermo Fisher Scientific). Soluble CD1c proteins were purified using nickel affinity and size exclusion chromatography. For CD1c-endo tetramers, purified CD1c-endo monomers were used to generate fluorescently labelled CD1 tetramers by conjugation to PE-streptavidin (BioLegend). To generate dextramers, the same process was repeated as above, with PE-labelled dextran backbones (Dex-PE) (Immudex).

TCR sequencing

Single CD1c-endo tetramer-positive cells were isolated by flow cytometric sorting into individual wells of a 96-well plate. Cells were lysed, and an oligo-dT primer was annealed to the poly-A tails of mRNA before reverse transcription to generate cDNA using an approach adapted from the SMART-seq2 protocol (Picelli *et al.*, 2014). During reverse transcription, adaptor sequences were incorporated into the cDNA, enabling subsequent universal PCR amplification using primers targeting the adaptor regions. Targeted PCRs were then performed for each single cell to amplify TCR variable regions using reverse primers specific for TRAC, TRBC, TRGC, and TRDC constant regions. PCR products were barcoded with an 8 bp DNA sequence unique to each well. PCR products were purified, and DNA fragments of 500–600 bp were size-selected. Next-generation sequencing libraries were prepared from size-selected fragments using New England Biolabs library preparation kits and sequenced

on the MiSeq platform. Sequences covering CDR3, variable, and constant regions were obtained and analysed using Seven Bridges software to generate final TCR sequences. Primer sequences are provided in **Supplementary file 1**.

TCR cloning and transduction

Sequences identified from tetramer-guided sorting were constructed into full-length TCR, synthesised, and sub-cloned into the pELNS lentivector by GenScript. Adherent HEK 293TN cells were co-transfected with pELNS lentivector (2.5 µg) and three accessory plasmids: pCMV-VSV-G (1.5 µg), pRSV.REV (3 µg), and pMDL.pg.RRE (3 µg), to induce production of lentiviral particles. Lentiviral particles were harvested, filtered, and used directly for transduction of β2m knockout J.RT3-T3.5 Jurkat cells (kind gift from Pierre Vantourout, King's College London). Transduced cells were sorted by flow cytometry on a FACSAria IIU (BD Biosciences).

CD1c-endo streptamers

Strep-Tactin magnetic microbeads (IBA Lifesciences) were prepared by washing and resuspending in binding buffer (PBS, 1 mM EDTA, 0.5% bovine serum albumin) prior to mixing with CD1c-endo incorporating Strep•Tag II peptide sequence and incubated at 4°C overnight. The following day, additional CD1c-endo was added and incubated for 30 min at 4°C to complete the preparation of the CD1c-endo streptamer. To enrich CD1c-restricted T-cells, 2×10^7 T-cells were resuspended in binding buffer containing 50 nM dasatinib (Axon) containing Human TruStain FcX (BioLegend) and incubated at room temperature for 10 min. CD1c-endo streptamer was then added to T-cells and incubated on ice for 20 min before separating the CD1c-endo streptamer-bound cells (positive fraction) from unbound cells (negative fraction) by magnetic sorting. The enrichment procedure was repeated with the negative fraction. The positive fractions were next pooled and thoroughly washed. To remove bound T-cells in the positive fraction from the CD1c-endo streptamer, the T-cells were resuspended in biotin elution buffer and incubated on ice for 10 min and separated using a magnet three times. The positive and negative fractions were stained and analysed by flow cytometry. CD1c-endo dextramer stained cells from the positive fraction were sorted by flow cytometry into a single well of a 96-well plate containing T-cell media and expanded with 1 µg/mL PHA, 100 IU/mL rhIL-2, and 5×10^5 irradiated PBMCs from three different donors.

Mtb culture

Mtb H37Rv (Mtb H37Rv) was cultured in Middlebrook 7H9 medium (supplemented with 10% ADC, 0.2% glycerol, and 0.02% Tween 80) (BD Biosciences) at 37°C in an incubator with shaking at 200 rpm. Bioluminescent Mtb H37Rv lux was cultured with kanamycin (25 µg/mL). Cultures at $\sim 1 \times 10^8$ CFU/mL Mtb (OD = 0.6) were used for all experiments.

Use of UV-killed and live Mtb

UV-killed and live Mtb were used for distinct experimental purposes. UV-killed Mtb was used where controlled exposure to defined amounts of Mtb-antigenic stimuli was required without ongoing bacterial replication, particularly for dose-response cytotoxicity and cytokine-release assays. Live Mtb infection was used to test whether responses observed with UV-killed Mtb were also evident during infection of APCs and to assess T-cell activation, target-cell lysis, and relative bacterial burden in the context of active host-pathogen interaction.

MoDC generation

To generate MoDC, CD14⁺ monocytes were isolated from PBMCs by negative selection using magnetic beads (STEMCELL Technologies) and cultured in the presence of 25 ng/mL rhGM-CSF and 20 ng/mL rhIL-4 (both from Miltenyi) for 5 days.

CD1 expression

MoDC

After 3 days of culturing monocytes in the presence of rhGM-CSF and rhIL-4, some wells were infected with live Mtb (MOI = 1) whereas others were left uninfected. On days 0, 3, 5, and 7, cells were

collected from wells by thoroughly washing with 3 mM EDTA and stained for flow cytometry analysis. Cells were acquired on FACS Aria IIU (BD Biosciences) to measure CD1a, CD1b, and CD1c expression.

THP1 cells

THP1-KO and THP1-CD1c cells were stained with anti-CD1c-PE and its isotype control (BioLegend) and Live/Dead Aqua (Invitrogen). On day 0, some wells were infected with Mtb (MOI = 1) whereas others were left uninfected. On day 3, cells were collected from wells by thoroughly washing with 3 mM EDTA and stained for flow cytometry analysis. Cells were acquired on FACS Aria IIU (BD Biosciences) to confirm the expression of CD1c on THP1 cells.

Immunohistochemistry

Tissue sections (4 µm thick) were dewaxed and rehydrated before inhibiting endogenous peroxidase. Heat-induced epitope retrieval was performed prior to blocking non-specific staining. Primary antibody, 1:500 anti-CD1c recombinant rabbit monoclonal (EPR23189-196, Abcam), was then applied to the sections and incubated overnight at 4°C. Sections incubated with buffer alone were included as a negative control. Secondary antibody, goat anti-rabbit 1:800 (2B Scientific), was applied. Sections were developed with avidin-biotin peroxidase complexes (2B Scientific) and 3,3'-diaminobenzidine tetrahydrochloride (Launch Diagnostics). Sections were counterstained with Mayer's haematoxylin, dehydrated, cleared, and mounted using XTF mounting medium (CellPath). Sections were allowed to dry before subsequent imaging using an Olympus Bx51 CC12 DotSlide microscope.

Differential gene expression analysis

PRJNA478394 FASTQ files were downloaded from ENA using enaBrowserTools (v1.1.0), after which transcripts in each file were quantified against a human genome template (downloaded from ftp://ftp.ensembl.org/pub/release106/fasta/homo_sapiens/cdna/Homo_sapiens.GRCh38.cdna.all.fa.gz) using alignment-free software kallisto (v0.46.1). Counts were imported into R using tximport (v1.28.0) and converted to counts per million (CPM) using edgeR (v3.14.0). Genes with summed CPM values below 14 across five samples (corresponding to the number of donors) were removed from the expression matrix, which was then normalised using TMM normalisation. Differential gene expression analysis was performed using the voom-limma (limma v3.56.2) pipeline, where normalised counts were compared between (live) Mtb-infected MoDCs and uninfected MoDCs (controls) at 48 hr. Computed p-values were adjusted for multiple hypothesis testing using the Benjamini-Hochberg method, and differentially expressed genes ($>+1 \log_2\text{FC}$ = upregulated, $<-1 \log_2\text{FC}$ = downregulated with an adjusted p-value <0.05) were subsequently identified.

Gene expression heatmap

Counts in the filtered, normalised expression matrix were converted to CPM, then $\log_2$ transformed. The expression matrix was subset to retain values for *CD1A*, *CD1B*, and *CD1C* for the control and (live) Mtb-infected samples at 48 hr post-infection. Coolmap function from limma was used to generate a heatmap showing the relative change in gene expression for each sample, from the average expression for each gene, using 'de pattern' option. Columns were clustered using 'average' method.

Activation assays

T-cells

T-cells were cultured with THP1-KO and THP1-CD1c cells that were either uninfected or infected with live Mtb (MOI = 1) for 48 hr at 37°C. Activation status of T-cells was measured through CD69 and CD25 expression by flow cytometry on a FACS Aria IIU (BD Biosciences).

Jurkat T-cells

TCR-transduced J.RT3-T3.5 Jurkat T-cells were cultured for 20 hr at 37°C in 96-well plates coated with CD1c-endo protein monomers (10 µg/mL). TCR-transduced J.RT3-T3.5 Jurkat T-cells cultured alone served as negative control. TCR-transduced J.RT3-T3.5 Jurkat T-cells were also cultured with

THP1-KO and THP1-CD1c cells that were either uninfected or infected with live Mtb (MOI = 1) for 20 hr at 37°C. Activation status of Jurkat T-cells was measured through CD69 expression by flow cytometry on a FACS Aria IIU.

AIM assay

Pan T-cells were isolated from healthy human blood using an EasySep Human T-cell Isolation Kit (STEMCELL Technologies) and stained with CTV to track proliferation. Purified T-cells are stimulated with THP1-CD1c cells for 12 days as above. On day 12, T-cells were re-stimulated overnight with irradiated THP1-KO, THP1-CD1c, or left unstimulated. Proliferated T-cells were determined as CTV low, and activation was measured as upregulation of CD25⁺/CD137⁺ or CD69⁺/CD137⁺ on T-cells by flow cytometry.

Cytotoxicity assays

ToxiLight cytotoxicity assay

T-cell cytotoxicity against UV-killed Mtb-treated target cells was measured using a ToxiLight assay (Lonza), which was carried out as per the manufacturer's instructions. Briefly, 20 µL of media from each well was transferred to a luminometer-compatible 96-well plate, 100 µL of ToxiLight reagent was then added. Following a 5 min incubation period, the plate was read using a Glomax 20/20 Luminometer (Glo-Max Discover; Promega).

Lysis assay

Tag-it Violet (BioLegend) stained 5×10⁴ THP1-KO or THP1-CD1c cells were left either uninfected or infected with live Mtb (MOI = 1) or stimulated with either 1 µg/mL LPS (Sigma) or 1 µg/mL Pam3CSK4 (InvivoGen) for 48 hr. These cells were then co-cultured with 2.5×10³ CD1c-autoreactive T-cells or cultured alone as matched APC-only controls for 48 hr. Cells were collected, stained with fixable live/dead stain, and analysed by flow cytometry. THP1 target cells were identified as Tag-it Violet-positive cells, and the frequency of dead target cells was determined within this gate. Specific lysis was calculated as the percentage of dead THP1 cells in the T-cell co-culture condition minus the percentage of dead THP1 cells in the matched APC-only control. This normalisation was performed independently for each matched condition within each experiment.

Luminex xMAP assays

T-cells were re-stimulated overnight with irradiated THP1-KO or THP1-CD1c cells, or left unstimulated. Additionally, CD1c-autoreactive T-cells, or control irrelevant T-cells, were co-cultured with untreated or UV-killed Mtb (MOI 10) treated THP1-KO and THP1-CD1c cells. Mtb untreated or treated THP1 cells and T-cells were cultured alone as control. At 24 hr after stimulation, supernatants were removed from each T-cell co-culture for multiplex cytokine analysis, cells were pelleted, and the supernatants were frozen. We followed the manufacturer's protocol to determine concentrations of cytokines (Life Technologies). A Bioplex 200 platform (Bio-Rad) was used to measure concentrations of cytokines in a multiplex panel: GM-CSF, IFN-γ, IL-1β, IL-10, IL-12p70, IL-13, IL-17a, IL-2, IL-22, IL-4, IL-5, IL-8, IP-10, MIP-1α, MIP-1β, Oncostatin M, RANTES, and TNF-α or the cytokine 35-plex human panel (Thermo Fisher Scientific).

Mtb Lux growth assays

THP1-KO and THP1-CD1c were infected with Mtb lux (MOI = 1). After overnight infection, cells were transferred from vented flasks to 50 mL Falcon tubes after detachment with Versene solution (Sigma-Aldrich) for 10 min. Cells were then washed with Hanks' balanced salt solution (HBSS) without Ca/Mg (Thermo Fisher Scientific) and then centrifuged at 380×g for 5 min at 4°C to remove extracellular bacteria. THP1-KO and THP1-CD1c were next resuspended in T-cell media containing kanamycin (25 µg/mL), co-cultured with T-cells and incubated at 37°C, 21% O₂, and 5% CO₂. Mtb growth was monitored by luminescence (GloMax 20/20 Luminometer, Promega) for 7 days. Luminescence values reflect viable lux-expressing Mtb under the assay conditions and do not distinguish intracellular from extracellular bacteria. Relative luminescence values for each condition were normalised to the matched infected APC-only control within each experimental replicate, allowing comparison of net bacterial burden across co-culture conditions.

Single-cell library preparation and sequencing

CD3-enriched PBMCs were washed in cold staining buffer (1% BSA in PBS), counted, and assessed for viability. One million cells per sample were resuspended in binding buffer containing 50 nM dasatinib (Axon), blocked (as above) and then stained with PE-dCODE dextramers conjugated to CD1c-endo, in combination with the following fluorochrome-conjugated antibodies: FITC-conjugated anti-human CD3 (clone UCHT1) and viability dye (Live/Dead Near IR, Invitrogen, Basingstoke, UK). Following a 30 min incubation at 4°C, cells were washed twice with ice-cold PBS supplemented with 1% FBS and sorted on a BD FACSAria cell sorter (BD Biosciences). Sorted cell fractions were labelled with sample-specific TotalSeq-C hashtag antibodies. Flow cytometry data were analysed using FlowJo v10.8.1 (FlowJo LLC). Sorted cells were loaded onto a 10x Genomics Chromium Chip G at a target of ~45,000 cells per lane using the Single Cell V(D)J Reagent Kit v1.1. First-strand cDNA synthesis was performed according to the manufacturer's protocol. After emulsion disruption and nucleic acid recovery, cDNA amplification proceeded following the Chromium Next GEM Single Cell V(D)J v1.1 user guide.

scRNA-seq alignment and analysis of CD1c-endo dextramer-sorted T-cells

Sequencing reads were de-multiplexed and aligned using Cell Ranger v7.2.0, with mapping to the GRCh38 reference genome (build 2020A, 10x Genomics). A total of 14,524 single CD3⁺ T-cells were initially profiled by scRNA-seq following dextramer-guided cell sorting. Cells were sorted into CD3⁺CD1c-endo-positive and CD3⁺CD1c-endo-negative fractions using dCODE dextramer reagents. Raw gene expression matrices were generated and processed for downstream analysis.

Quality control and filtering

Cells underwent rigorous quality control to ensure data integrity and remove low-quality or ambiguous profiles. Specifically, the following metrics were calculated per cell: number of detected genes, total unique molecular identifiers (UMIs), mitochondrial gene content, and ribosomal gene content. Additionally, two independent doublet detection methods were applied: Scrublet and scDblFinder. Outlier cells were excluded based on the following criteria: low gene content and UMI counts, high mitochondrial content (>10%), high ribosomal content (>50%), and predicted doublets from at least one of the two detection tools. After filtering, a total of 11,804 high-quality single T-cells were retained for downstream analysis.

Annotation of CD1c-endo binding

Cells were annotated as CD1c-endo-positive or -negative based on dextramer binding. Of the 11,804 that passed quality control, 9518 lacked detectable dextramer binding and were initially labelled as CD1c-endo negative. These were further stratified by sorting origin: 7652 cells originated from the CD1c-endo-positive sort gate and 1866 cells came from the CD1c-endo-negative sort gate. Only the 1866 cells from the CD1c-endo-negative sort gate were retained as confidently assigned CD1c-endo-negative T-cells. The remaining 7652 cells, despite lacking dextramer signal, were excluded due to possible false negatives introduced during staining or sorting. To accurately classify CD1c-endo-positive cells, a density-based thresholding approach was applied. The distribution of dextramer signal was modelled using kernel density estimation (KDE). To define confidently positive cells, thresholds were set based on the 1st and 99th percentiles and the interquartile range (IQR) of dextramer intensity. Cells with signal below the lower bound ($Q1 - 1.5 \times IQR$) were excluded as potential noise or non-specific binders. This approach minimised false positives and enabled robust separation of CD1c-endo-positive and -negative populations. After applying quality control filters and dextramer-based gating, CD1c-endo-positive T-cells were defined as those confidently binding dextramer above the KDE threshold, while CD1c-endo-negative T-cells comprised the 1866 cells from the negative gate lacking any detectable dextramer signal.

Subsampling analysis to assess robustness

To control for group size imbalance (CD1c-endo⁺ vs CD1c-endo⁻), we performed a subsampling analysis. From each group, a random subset of cells equal to the smallest group size (1866) was sampled

using `np.random.choice` (without replacement). This balanced dataset was reanalysed to ensure that key differential gene expression patterns and clustering results were consistent across equal-sized groups. The findings from the full dataset were confirmed, indicating that group size imbalance did not bias the results, validating the robustness of the conclusions.

Study approval

All clinical studies were conducted according to the Declaration of Helsinki principles. All participants gave written informed consent. The participant study was approved by NRES Committee South Central – Southampton A (REC 13/SC/0043; IRAS 109038). Paraffin-embedded lung tissue from TB patients or those with adenocarcinoma were retrieved from the histology archive at University Hospital Southampton with approval by the Institutional Review Board (Reference 12/NW/0794: SRB04_14). The ethics committee approved immunohistochemical analysis without individual informed consent since it was surplus archived tissue taken as part of routine care.

Statistical analysis

GraphPad Prism version 10 (GraphPad Software, Inc) was used for statistical analysis, and p -values ≤ 0.05 were considered statistically significant. The Mann-Whitney U test, t tests, one-way ANOVA, Wilcoxon signed-rank test, two-way ANOVA, and Tukey's test were used as stated in the figure legends.

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Additional information

Competing interests

Rita Szoke-Kovacs, Marco Lepore, David K Cole: Employee of Immunocore Limited. The other authors declare that no competing interests exist.

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Author contributions

Matthew Milton, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft; Sahar H Farag, Kinga Niedobecka, Richard Stopforth, Investigation; Diana Garay-Baquero, Jennie Gullick, Data curation, Formal analysis, Investigation, Methodology; Daniel Burns, Laura Denney, Investigation, Methodology; Rita Szoke-Kovacs, Data curation, Investigation, Methodology; Patrick Trimby-Smith, Data curation, Formal analysis, Investigation; Alex Look, Formal analysis, Investigation; Marco Lepore, David K Cole, Supervision; Andrew White, Supervision, Funding acquisition; Sally Sharpe, Funding acquisition, Investigation; Alasdair Leslie, Supervision, Investigation, Writing – review and editing; Andres Vallejo, Supervision, Investigation, Methodology; Liku Tezera, Supervision, Investigation, Methodology, Writing – review and editing; Paul Elkington, Supervision, Funding acquisition, Investigation, Methodology, Writing – original draft, Writing – review and editing; Salah Mansour, Conceptualization, Formal analysis, Supervision, Funding acquisition, Investigation, Writing – original draft, Project administration, Writing – review and editing

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Ethics

The study involved human samples and was conducted in accordance with the principles of the Declaration of Helsinki. Written informed consent was obtained where required. Paraffin-embedded lung tissue from patients with tuberculosis was obtained from the histology archive at University Hospital Southampton with Institutional Review Board approval (Reference 12/NW/0794: SRB04_14). The ethics committee approved use of surplus archived tissue obtained during routine clinical care without individual informed consent. Peripheral blood mononuclear cells were obtained from anonymised healthy blood-bank donors. No identifiable participant information is reported.

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Additional files

Supplementary files

Supplementary file 1. Primer sequences used for targeted single-cell TCR sequencing. Sequences are shown for the oligo-dT primer, universal forward and reverse primers, targeted forward primer containing the 8 bp well barcode, and reverse primers specific for TRAC, TRBC, TRGC, and TRDC constant regions.

MDAR checklist

Source data 1. Numerical source data for **Figures 1–6** and associated figure supplements. Data are organised by figure panel in separate worksheets.

Data availability

Source data files are provided with this article. scRNA-seq data are available from the European Nucleotide Archive under accession PRJEB94332. The previously published bulk RNA-seq dataset analysed in this study is available under accession PRJNA478394. Newly generated cell lines and TCR constructs are available from the corresponding author upon reasonable request.

The following datasets were generated:

Author(s)	Year	Dataset title	Dataset URL	Database and Identifier
Pacis A, Mailhot-Léonard F, Randolph HE, Yotova V, Dumaine A, Grenier J-C, Barreiro LB	2026	Gene activation precedes DNA demethylation in response to infection in human dendritic cells	https://seqout.org/p/PRJNA478394	Seqout, PRJNA478394
Milton M, Farag S, Garay-Baquero DJ, Gullick J, Niedobbecka K, Burns D, Szoke-Kovacs R, Trimby-Smith P, Look A, Stopforth RJ, Lepore M, Cole DK, Denney L, White A, Sharpe S, Leslie A, Vallejo A, Tezera LB, Elkington PT, Mansour S	2025	sc-RNAseq on CD1c restricted T-cells	https://www.ebi.ac.uk/ena/browser/view/PRJEB94332	European Nucleotide Archive, PRJEB94332

References

- Aquino A**, Graziani G, Franzese O, Prete SP, Bonmassar E, Bonmassar L, D'Atri S. 2011. Exogenous control of the expression of Group I CD1 molecules competent for presentation of microbial nonpeptide antigens to human T lymphocytes. *Clinical & Developmental Immunology* **2011**:790460. DOI: <https://doi.org/10.1155/2011/790460>, PMID: 21603161
- Bielecka MK**, Tezera LB, Zmijan R, Drobniowski F, Zhang X, Jayasinghe S, Elkington P. 2017. A bioengineered three-dimensional cell culture platform integrated with microfluidics to address antimicrobial resistance in tuberculosis. *mBio* **8**:e02073-16. DOI: <https://doi.org/10.1128/mBio.02073-16>, PMID: 28174307
- Brennan PJ**, Tatituri RVV, Brigl M, Kim EY, Tuli A, Sanderson JP, Gadola SD, Hsu F-F, Besra GS, Brenner MB. 2011. Invariant natural killer T cells recognize lipid self antigen induced by microbial danger signals. *Nature Immunology* **12**:1202–1211. DOI: <https://doi.org/10.1038/ni.2143>, PMID: 22037601
- Brites D**, Gagneux S. 2015. Co-evolution of *Mycobacterium tuberculosis* and *Homo sapiens*. *Immunological Reviews* **264**:6–24. DOI: <https://doi.org/10.1111/imr.12264>, PMID: 25703549
- Chancellor A**, Gadola SD, Mansour S. 2018. The versatility of the CD1 lipid antigen presentation pathway. *Immunology* **154**:196–203. DOI: <https://doi.org/10.1111/imm.12912>, PMID: 29460282

- Comas I, Coscolla M, Luo T, Borrell S, Holt KE, Kato-Maeda M, Parkhill J, Malla B, Berg S, Thwaites G, Yeboah-Manu D, Bothamley G, Mei J, Wei L, Bentley S, Harris SR, Niemann S, Diel R, Aseffa A, Gao Q, et al. 2013. Out-of-Africa migration and Neolithic coexpansion of *Mycobacterium tuberculosis* with modern humans. *Nature Genetics* **45**:1176–1182. DOI: <https://doi.org/10.1038/ng.2744>, PMID: 23995134
- Cotton RN, Shahine A, Rossjohn J, Moody DB. 2018. Lipids hide or step aside for CD1-autoreactive T cell receptors. *Current Opinion in Immunology* **52**:93–99. DOI: <https://doi.org/10.1016/j.coi.2018.04.013>, PMID: 29738961
- de Jong A, Arce EC, Cheng TY, van Summeren RP, Feringa BL, Dudkin V, Crich D, Matsunaga I, Minnaard AJ, Moody DB. 2007. CD1c presentation of synthetic glycolipid antigens with foreign Alkyl Branching Motifs. *Chemistry & Biology* **14**:1232–1242. DOI: <https://doi.org/10.1016/j.chembiol.2007.09.010>, PMID: 18022562
- de Jong A, Peña-Cruz V, Cheng T-Y, Clark RA, Van Rhijn I, Moody DB. 2010. CD1a-autoreactive T cells are a normal component of the human $\alpha\beta$ T cell repertoire. *Nature Immunology* **11**:1102–1109. DOI: <https://doi.org/10.1038/ni.1956>, PMID: 21037579
- de Lalla C, Lepore M, Piccolo FM, Rinaldi A, Scelfo A, Garavaglia C, Mori L, De Libero G, Dellabona P, Casorati G. 2011. High-frequency and adaptive-like dynamics of human CD1 self-reactive T cells. *European Journal of Immunology* **41**:602–610. DOI: <https://doi.org/10.1002/eji.201041211>, PMID: 21246542
- Felio K, Nguyen H, Dascher CC, Choi H-J, Li S, Zimmer MI, Colmone A, Moody DB, Brenner MB, Wang C-R. 2009. CD1-restricted adaptive immune responses to *Mycobacteria* in human group 1 CD1 transgenic mice. *The Journal of Experimental Medicine* **206**:2497–2509. DOI: <https://doi.org/10.1084/jem.20090898>, PMID: 19808251
- Fumagalli M, Sironi M, Pozzoli U, Ferrer-Admetlla A, Pattini L, Nielsen R. 2011. Signatures of environmental genetic adaptation pinpoint pathogens as the main selective pressure through human evolution. *PLOS Genetics* **7**:e1002355. DOI: <https://doi.org/10.1371/journal.pgen.1002355>, PMID: 22072984
- Gagliardi MC, Teloni R, Mariotti S, Iona E, Pardini M, Fattorini L, Orefici G, Nisini R. 2004. *Bacillus Calmette-Guérin* shares with virulent *Mycobacterium tuberculosis* the capacity to subvert monocyte differentiation into dendritic cell: implication for its efficacy as a vaccine preventing tuberculosis. *Vaccine* **22**:3848–3857. DOI: <https://doi.org/10.1016/j.vaccine.2004.07.009>, PMID: 15364431
- Gherardin NA, Redmond SJ, McWilliam HEG, Almeida CF, Gourley KHA, Seneviratna R, Li S, De Rose R, Ross FJ, Nguyen-Robertson CV, Su S, Ritchie ME, Villadangos JA, Moody DB, Pellicci DG, Uldrich AP, Godfrey DI. 2021. CD36 family members are TCR-independent ligands for CD1 antigen-presenting molecules. *Science Immunology* **6**:eabg4176. DOI: <https://doi.org/10.1126/sciimmunol.abg4176>, PMID: 34172588
- Grailler J, Cheng ZJ, Hartnett J, Slater M, Fan F, Cong M. 2023. A novel cell-based luciferase reporter platform for the development and characterization of T-cell redirecting therapies and vaccine development. *Journal of Immunotherapy* **46**:96–106. DOI: <https://doi.org/10.1097/CJI.0000000000000453>, PMID: 36809225
- Guo T, Koo MY, Kagoya Y, Anczurowski M, Wang CH, Saso K, Butler MO, Hirano N. 2018. A subset of human autoreactive CD1c-Restricted T cells preferentially expresses TRBV4-1* TCRs. *The Journal of Immunology* **200**:500–511. DOI: <https://doi.org/10.4049/jimmunol.1700677>, PMID: 29237773
- Huang S, Shahine A, Cheng TY, Chen YL, Ng SW, Balaji GR, Farquhar R, Gras S, Hardman CS, Altman JD, Tahiri N, Minnaard AJ, Ogg GS, Mayfield JA, Rossjohn J, Moody DB. 2023. CD1 lipidomes reveal lipid-binding motifs and size-based antigen-display mechanisms. *Cell* **186**:4583–4596. DOI: <https://doi.org/10.1016/j.cell.2023.08.022>, PMID: 37725977
- Hui D, Sahu S, Ditiu L, Marais BJ, Yeboah-Manu D, Goletti D, Walzl G, Zumla A. 2026. Global tuberculosis response off track: urgent priorities to end the world's top infectious killer. *The Lancet* **407**:1126–1129. DOI: [https://doi.org/10.1016/S0140-6736\(25\)02433-X](https://doi.org/10.1016/S0140-6736(25)02433-X)
- Jasenovsky LD, Scriba TJ, Hanekom WA, Goldfeld AE. 2015. T cells and adaptive immunity to *Mycobacterium tuberculosis* in humans. *Immunological Reviews* **264**:74–87. DOI: <https://doi.org/10.1111/immr.12274>, PMID: 25703553
- Joosten SA, Ottenhoff THM, Lewinsohn DM, Hoft DF, Moody DB, Seshadri C, Collaboration for Tuberculosis Vaccine Discovery - Donor-Unrestricted T-cells Working Group, Bill and Melinda Gates Foundation. 2019. Harnessing donor unrestricted T-cells for new vaccines against tuberculosis. *Vaccine* **37**:3022–3030. DOI: <https://doi.org/10.1016/j.vaccine.2019.04.050>, PMID: 31040086
- Karlsson EK, Kwiatkowski DP, Sabeti PC. 2014. Natural selection and infectious disease in human populations. *Nature Reviews. Genetics* **15**:379–393. DOI: <https://doi.org/10.1038/nrg3734>, PMID: 24776769
- Lepore M, de Lalla C, Gundimeda SR, Gsellinger H, Consonni M, Garavaglia C, Sansano S, Piccolo F, Scelfo A, Häussinger D, Montagna D, Locatelli F, Bonini C, Bondanza A, Forcina A, Li Z, Ni G, Ciceri F, Jenö P, Xia C, et al. 2014. A novel self-lipid antigen targets human T cells against CD1c(+) leukemias. *The Journal of Experimental Medicine* **211**:1363–1377. DOI: <https://doi.org/10.1084/jem.20140410>, PMID: 24935257
- Li S, Choi HJ, Felio K, Wang CR. 2011. Autoreactive CD1b-restricted T cells: a new innate-like T-cell population that contributes to immunity against infection. *Blood* **118**:3870–3878. DOI: <https://doi.org/10.1182/blood-2011-03-341941>, PMID: 21860021
- Ly D, Kasmar AG, Cheng T-Y, de Jong A, Huang S, Roy S, Bhatt A, van Summeren RP, Altman JD, Jacobs WR Jr, Adams EJ, Minnaard AJ, Porcelli SA, Moody DB. 2013. CD1c tetramers detect ex vivo T cell responses to processed phosphomycoketide antigens. *The Journal of Experimental Medicine* **210**:729–741. DOI: <https://doi.org/10.1084/jem.20120624>, PMID: 23530121
- Mansour S, Tocheva AS, Cave-Ayland C, Machelett MM, Sander B, Lissin NM, Molloy PE, Baird MS, Stübs G, Schröder NWJ, Schumann RR, Rademann J, Postle AD, Jakobsen BK, Marshall BG, Gosain R, Elkington PT,

- Elliott T, Skylaris C-K, Essex JW, et al. 2016. Cholesteryl esters stabilize human CD1c conformations for recognition by self-reactive T cells. *PNAS* **113**:E1266–E1275. DOI: <https://doi.org/10.1073/pnas.1519246113>, PMID: 26884207
- Matsunaga I, Bhatt A, Young DC, Cheng TY, Eyles SJ, Besra GS, Briken V, Porcelli SA, Costello CE, Jacobs WR Jr, Moody DB. 2004. *Mycobacterium tuberculosis* pks12 produces a novel polyketide presented by CD1c to T cells. *The Journal of Experimental Medicine* **200**:1559–1569. DOI: <https://doi.org/10.1084/jem.20041429>, PMID: 15611286
- Melián A, Geng YJ, Sukhova GK, Libby P, Porcelli SA. 1999. CD1 expression in human atherosclerosis. A potential mechanism for T cell activation by foam cells. *The American Journal of Pathology* **155**:775–786. DOI: [https://doi.org/10.1016/S0002-9440\(10\)65176-0](https://doi.org/10.1016/S0002-9440(10)65176-0), PMID: 10487835
- Moody DB, Ulrichs T, Mühlecker W, Young DC, Gurucha SS, Grant E, Rosat JP, Brenner MB, Costello CE, Besra GS, Porcelli SA. 2000. CD1c-mediated T-cell recognition of isoprenoid glycolipids in *Mycobacterium tuberculosis* infection. *Nature* **404**:884–888. DOI: <https://doi.org/10.1038/35009119>, PMID: 10786796
- Mori L, Lepore M, De Libero G. 2016. The immunology of CD1- and MR1-restricted T cells. *Annual Review of Immunology* **34**:479–510. DOI: <https://doi.org/10.1146/annurev-immunol-032414-112008>, PMID: 26927205
- Ndiaye BP, Thienemann F, Ota M, Landry BS, Camara M, Dièye S, Dieye TN, Esmail H, Goliath R, Huygen K, January V, Ndiaye I, Oni T, Raine M, Romano M, Satti I, Sutton S, Thiam A, Wilkinson KA, Mboup S, et al. 2015. Safety, immunogenicity, and efficacy of the candidate tuberculosis vaccine MVA85A in healthy adults infected with HIV-1: a randomised, placebo-controlled, phase 2 trial. *The Lancet. Respiratory Medicine* **3**:190–200. DOI: [https://doi.org/10.1016/S2213-2600\(15\)00037-5](https://doi.org/10.1016/S2213-2600(15)00037-5), PMID: 25726088
- Nguipod-Djomo P, Haldal E, Rodrigues LC, Abubakar I, Mangtani P. 2016. Duration of BCG protection against tuberculosis and change in effectiveness with time since vaccination in Norway: a retrospective population-based cohort study. *The Lancet. Infectious Diseases* **16**:219–226. DOI: [https://doi.org/10.1016/S1473-3099\(15\)00400-4](https://doi.org/10.1016/S1473-3099(15)00400-4), PMID: 26603173
- Nunes-Alves C, Booty MG, Carpenter SM, Jayaraman P, Rothchild AC, Behar SM. 2014. In search of a new paradigm for protective immunity to TB. *Nature Reviews. Microbiology* **12**:289–299. DOI: <https://doi.org/10.1038/nrmicro3230>, PMID: 24590243
- O’Garra A, Redford PS, McNab FW, Bloom CI, Wilkinson RJ, Berry MPR. 2013. The immune response in tuberculosis. *Annual Review of Immunology* **31**:475–527. DOI: <https://doi.org/10.1146/annurev-immunol-032712-095939>, PMID: 23516984
- Pacis A, Mailhot-Léonard F, Tailleux L, Randolph HE, Yotova V, Dumaine A, Grenier JC, Barreiro LB. 2019. Gene activation precedes DNA demethylation in response to infection in human dendritic cells. *PNAS* **116**:6938–6943. DOI: <https://doi.org/10.1073/pnas.1814700116>, PMID: 30886108
- Paulson T. 2013. Epidemiology: A mortal foe. *Nature* **502**:S2–S3. DOI: <https://doi.org/10.1038/502S2a>, PMID: 24108078
- Picelli S, Faridani OR, Björklund AK, Winberg G, Sagasser S, Sandberg R. 2014. Full-length RNA-seq from single cells using Smart-seq2. *Nature Protocols* **9**:171–181. DOI: <https://doi.org/10.1038/nprot.2014.006>, PMID: 24385147
- Reichmann MT, Tezera LB, Denney L, Schiff H, Vallejo A, Mansour S, Leslie A, Garay-Baquero DJ, Elkington PT. 2026. A co-evolutionary perspective on humans and *Mycobacterium tuberculosis* in the era of systems biology. *eLife* **14**:RP108175. DOI: <https://doi.org/10.7554/eLife.108175>, PMID: 41499279
- Reinink P, Souter MNT, Cheng T-Y, van Gorkom T, Lenz S, Kubler-Kielb J, Strle K, Kremer K, Thijsen SFT, Steere AC, Godfrey DI, Pellicci DG, Moody DB, Van Rhijn I. 2019. CD1b presents self and *Borrelia burgdorferi* diacylglycerols to human T cells. *European Journal of Immunology* **49**:737–746. DOI: <https://doi.org/10.1002/eji.201847949>, PMID: 30854633
- Reljic R, Stylianou E, Balu S, Ma JK. 2010. Cytokine interactions that determine the outcome of *Mycobacterial* infection of macrophages. *Cytokine* **51**:42–46. DOI: <https://doi.org/10.1016/j.cyto.2010.04.005>, PMID: 20434357
- Roura-Mir C, Catálfamo M, Cheng T-Y, Marqusee E, Besra GS, Jaraquemada D, Moody DB. 2005. CD1a and CD1c activate intrathyroidal T cells during Graves’ disease and Hashimoto’s thyroiditis. *Journal of Immunology* **174**:3773–3780. DOI: <https://doi.org/10.4049/jimmunol.174.6.3773>, PMID: 15749918
- Roy S, Ly D, Li N-S, Altman JD, Piccirilli JA, Moody DB, Adams EJ. 2014. Molecular basis of mycobacterial lipid antigen presentation by CD1c and its recognition by $\alpha\beta$ T cells. *PNAS* **111**:E4648–E4657. DOI: <https://doi.org/10.1073/pnas.1408549111>, PMID: 25298532
- Roy S, Ly D, Castro CD, Li NS, Hawk AJ, Altman JD, Meredith SC, Piccirilli JA, Moody DB, Adams EJ. 2016. Molecular analysis of lipid-reactive V δ 1 $\gamma\delta$ T cells identified by CD1c tetramers. *Journal of Immunology* **196**:1933–1942. DOI: <https://doi.org/10.4049/jimmunol.1502202>, PMID: 26755823
- Ruibal P, Voogd L, Joosten SA, Ottenhoff THM. 2021. The role of donor-unrestricted T-cells, innate lymphoid cells, and NK cells in anti-mycobacterial immunity. *Immunological Reviews* **301**:30–47. DOI: <https://doi.org/10.1111/imr.12948>, PMID: 33529407
- Scharf L, Li N-S, Hawk AJ, Garzón D, Zhang T, Fox LM, Kazen AR, Shah S, Haddadian EJ, Gumperz JE, Saghatelian A, Faraldo-Gómez JD, Meredith SC, Piccirilli JA, Adams EJ. 2010. The 2.5 Å structure of CD1c in complex with a mycobacterial lipid reveals an open groove ideally suited for diverse antigen presentation. *Immunity* **33**:853–862. DOI: <https://doi.org/10.1016/j.immuni.2010.11.026>, PMID: 21167756

- Schrager LK**, Vekemens J, Drager N, Lewinsohn DM, Olesen OF. 2020. The status of tuberculosis vaccine development. *The Lancet. Infectious Diseases* **20**:e28–e37. DOI: [https://doi.org/10.1016/S1473-3099\(19\)30625-5](https://doi.org/10.1016/S1473-3099(19)30625-5), PMID: 32014117
- Steling PA**, Porcelli SA, Duong BT, Spada F, Bloom BR, Diamond B, Hahn BH. 2000. Human double-negative T cells in systemic lupus erythematosus provide help for IgG and are restricted by CD1c. *Journal of Immunology* **165**:5338–5344. DOI: <https://doi.org/10.4049/jimmunol.165.9.5338>, PMID: 11046069
- Spada FM**, Grant EP, Peters PJ, Sugita M, Melián A, Leslie DS, Lee HK, van Donselaar E, Hanson DA, Krensky AM, Majdic O, Porcelli SA, Morita CT, Brenner MB. 2000. Self-recognition of CD1 by gamma/delta T cells: implications for innate immunity. *The Journal of Experimental Medicine* **191**:937–948. DOI: <https://doi.org/10.1084/jem.191.6.937>, PMID: 10727456
- Stenger S**, Hanson DA, Teitelbaum R, Dewan P, Niazi KR, Froelich CJ, Ganz T, Thoma-Uszynski S, Melián A, Bogdan C, Porcelli SA, Bloom BR, Krensky AM, Modlin RL. 1998a. An antimicrobial activity of cytolytic T cells mediated by granulysin. *Science* **282**:121–125. DOI: <https://doi.org/10.1126/science.282.5386.121>, PMID: 9756476
- Stenger S**, Niazi KR, Modlin RL. 1998b. Down-regulation of CD1 on antigen-presenting cells by infection with *Mycobacterium tuberculosis*. *The Journal of Immunology* **161**:3582–3588. DOI: <https://doi.org/10.4049/jimmunol.161.7.3582>, PMID: 9759880
- Suzuki TA**, Fitzstevens JL, Schmidt VT, Enav H, Huus KE, Mbong Ngwese M, Griebhammer A, Pfeleiderer A, Adegbite BR, Zinsou JF, Esen M, Velavan TP, Adegnika AA, Song LH, Spector TD, Muehlbauer AL, Marchi N, Kang H, Maier L, Blekhman R, et al. 2022. Codiversification of gut microbiota with humans. *Science* **377**:1328–1332. DOI: <https://doi.org/10.1126/science.abm7759>, PMID: 36108023
- Tait DR**, Hatherill M, Van Der Meeren O, Ginsberg AM, Van Brakel E, Salaun B, Scriba TJ, Akite EJ, Ayles HM, Bollaerts A, Demoitié M-A, Diacon A, Evans TG, Gillard P, Hellström E, Innes JC, Lempicki M, Malahleha M, Martinson N, Mesia Vela D, et al. 2019. Final analysis of a trial of M72/AS01_E vaccine to prevent tuberculosis. *The New England Journal of Medicine* **381**:2429–2439. DOI: <https://doi.org/10.1056/NEJMoa1909953>, PMID: 31661198
- Tameris MD**, Hatherill M, Landry BS, Scriba TJ, Snowden MA, Lockhart S, Shea JE, McClain JB, Hussey GD, Hanekom WA, Mahomed H, McShane H, MVA85A 020 Trial Study Team. 2013. Safety and efficacy of MVA85A, a new tuberculosis vaccine, in infants previously vaccinated with BCG: a randomised, placebo-controlled phase 2b trial. *Lancet* **381**:1021–1028. DOI: [https://doi.org/10.1016/S0140-6736\(13\)60177-4](https://doi.org/10.1016/S0140-6736(13)60177-4), PMID: 23391465
- Van Rhijn I**, van Berlo T, Hilmenyuk T, Cheng T-Y, Wolf BJ, Tatituri RVV, Uldrich AP, Napolitani G, Cerundolo V, Altman JD, Willemsen P, Huang S, Rossjohn J, Besra GS, Brenner MB, Godfrey DI, Moody DB. 2016. Human autoreactive T cells recognize CD1b and phospholipids. *PNAS* **113**:380–385. DOI: <https://doi.org/10.1073/pnas.1520947112>, PMID: 26621732
- Vincent MS**, Xiong X, Grant EP, Peng W, Brenner MB. 2005. CD1a-, b-, and c-restricted TCRs recognize both self and foreign antigens. *Journal of Immunology* **175**:6344–6351. DOI: <https://doi.org/10.4049/jimmunol.175.10.6344>, PMID: 16272286
- Wallis RS**, Maeurer M, Mwaba P, Chakaya J, Rustumjee R, Migliori GB, Marais B, Schito M, Churchyard G, Swaminathan S, Hoelscher M, Zumla A. 2016. Tuberculosis--advances in development of new drugs, treatment

- regimens, host-directed therapies, and biomarkers. *The Lancet. Infectious Diseases* **16**:e34–e46. DOI: [https://doi.org/10.1016/S1473-3099\(16\)00070-0](https://doi.org/10.1016/S1473-3099(16)00070-0), PMID: 27036358
- Wen Q**, Zhou C, Xiong W, Su J, He J, Zhang S, Du X, Liu S, Wang J, Ma L. 2016. MiR-381-3p regulates the antigen-presenting capability of dendritic cells and represses antituberculosis cellular immune responses by targeting CD1c. *Journal of Immunology* **197**:580–589. DOI: <https://doi.org/10.4049/jimmunol.1500481>, PMID: 27296666
- World Health Organization**. 2021. Global tuberculosis report 2021. Geneva: World Health Organization.
- Wun KS**, Reijneveld JF, Cheng T-Y, Ladell K, Uldrich AP, Le Nours J, Miners KL, McLaren JE, Grant EJ, Haigh OL, Watkins TS, Suliman S, Iwany S, Jimenez J, Calderon R, Tamara KL, Leon SR, Murray MB, Mayfield JA, Altman JD, et al. 2018. T cell autoreactivity directed toward CD1c itself rather than toward carried self lipids. *Nature Immunology* **19**:397–406. DOI: <https://doi.org/10.1038/s41590-018-0065-7>, PMID: 29531339
- Zhao J**, Siddiqui S, Shang S, Bian Y, Bagchi S, He Y, Wang C-R. 2015. Mycolic acid-specific T cells protect against *Mycobacterium tuberculosis* infection in a humanized transgenic mouse model. *eLife* **4**:e08525. DOI: <https://doi.org/10.7554/eLife.08525>, PMID: 26652001